S3A: A Self-Supervised Saliency Analysis Framework for Hyperspectral Image

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***Abstract*— Most existing visual saliency analysis methods are meticulously crafted for natural RGB images, which typically lean upon foundational spatial visual cues, such as contrast, structure, and texture in scenes. However, these methods exhibit limitations in spectral saliency analysis due to the existence of numerous materials that may manifest identical RGB values while harboring disparate spectral characteristics. Consequently, it is an exigency to introduce the saliency method, especially for hyperspectral images (HSIs). In this article, we establish a set of hyperspectral saliency principles that incorporate both spectral and spatial attributes, and accordingly present a novel self-supervised saliency analysis (S3A) framework for HSIs. Note that our S3A is designed as a discriminant architecture composed of three interconnected components. More specifically, an unsupervised HSI representative patch sampling (RPS) module is designed to pick up some representative pixels for S3A training. Subsequently, we construct a classification-based patch spectral discrimination network (CPSD-Net) to evaluate the HSI saliency. Finally, a patch-to-global spatial diffusion (P2G-SD) module is constructed to diffuse the saliency from few supervised samples to the other HSI pixels. Moreover, to demonstrate the performance of our saliency analysis framework, we apply the HSI saliency to some classic downstream tasks including band selection (BS) and HSI classification. On several popular HSI datasets, the satisfactory quantization results fully verify the rationality and effectiveness of our S3A framework, in terms of entropy value and mean spectral divergence (MSD) of the selected bands in BS task, as well as the accuracy in the HSI classification task. The source code is publicly available at https://github.com/Lee-zl/S3A**

***Index Terms*— Band selection (BS), HSI classification, hyperspectral image (HSI), saliency analysis, self-supervised learning.**

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I. Introduction

HYPERSPECTRAL imaging sensors can capture hundreds of different wavelength details of the same spatial region simultaneously, which can accurately reflect the rich spectral response characteristics of targets [4], [5], [6]. In the past decades, hyperspectral imaging technology and equipment have been developed rapidly, and have been widely used in civil and military fields, such as land cover statistics [7], biomedical image processing [8], precision agriculture and animal husbandry [9], [10], and battlefield environment monitoring [11]. However, extracting valuable information from these detailed-rich hyperspectral images (HSIs) demands intricate and precise analytical methods. In this context, visual saliency analysis [12], [13], [14] has emerged as a pivotal approach capable of identifying regions of interest (ROIs) within an image and facilitating rapid comprehension of its content [15]. Nonetheless, the current prevailing techniques for visual saliency analysis mainly cater to RGB images [16], [17], thereby presenting a myriad of challenges when extending these methods seamlessly to HSIs. In particular, conventional saliency analysis methods tailored to RGB images invariably rely on their three color channels, while hyperspectral data generally contain hundreds of spectral bands, which leads to significant dimensional differences [18], [19], [20]. The disparity in dimensionality constitutes a significant obstacle for RGB-based saliency methods to capture the intricate spectral details and complex spectral interactions in HSIs [21], [22], [23], [24], [25]. Therefore, their efficacy and applicability often sharply decrease when ported to hyperspectral data analysis.

To facilitate an intuitive visual understanding of this problem, we take the Indian Pines (IP) dataset as an example, as shown in Fig. 1. In the realm of traditional visual saliency analysis, typical RGB-based saliency methods, such as ITTI [Fig. 1(h)] [1], FT [Fig. 1(i)] [2], and MBD [Fig. 1(j)] [3], generally employ either a three-channel or grayscale image such as Fig. 1(c) is used as input. These approaches explore foundational spatial visual attributes inherent to images, facilitating the discernment of salient information [26], [27], [28]. Nevertheless, when confronted with HSIs, these methods obtain poor saliency analysis performance due to their difficulty in utilizing the massive spectral details [29], [30], [31]. As shown in Fig. 1(a), HSI encompasses a high spectral dimension, and each pixel can be delineated by a distinct spectral vector [32].

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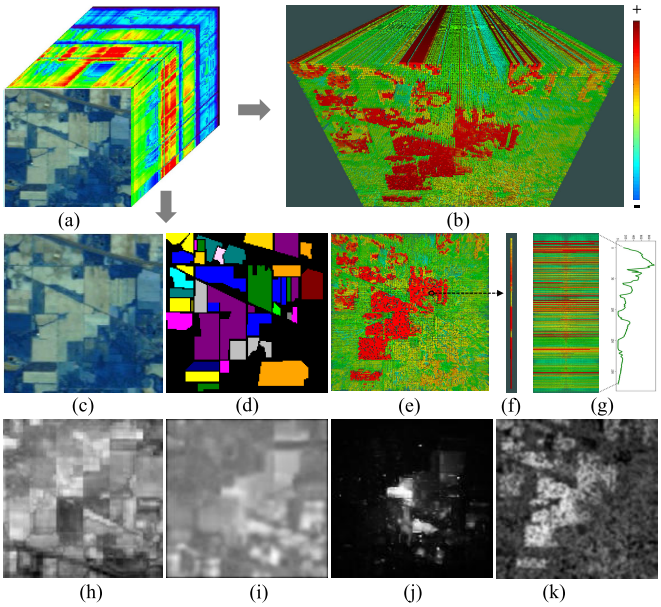


Fig. 1. Visualization of the saliency on the IP dataset. (a) Original HSI data. (b) HSI saliency cube produced by our S3A. (c) False-color image. (d) Land-cover category labels. (e) Spatial-domain saliency decomposed from the HSI saliency cube. (f) Spectralwise S3A curve of the marked pixel with a black circle in (e). (g) Spectral-domain saliency decomposed from the HSI saliency cube, and the standard deviation distribution corresponding to the average gray level of pixels between different categories in each band. The bottom row represents the saliency maps of false-color image in (c) via typical RGB image saliency methods including (h) ITTI [1], (i) FT [2], (j) MBD [3], and (k) ours (S3A).

In hyperspectral feature analysis, commonly used methods include band selection (BS) [33], [34], [35], band sampling [36], [37], and feature extraction (FE) [38], [39]. Although these approaches offer advantages in reducing data redundancy and computational complexity, they often struggle to effectively integrate spectral and spatial information, potentially leading to the loss of critical spectral details. Consequently, hyperspectral saliency analysis (HSA) has emerged as a promising alternative, garnering increasing attention in recent research. In previous work, HSA approaches [40], [41], [42], [43], [44], [45] can be roughly divided into two branches, feature-compared HSI saliency [43], [44] and task-driven HSI saliency [41], [45]. The former aims to locate the ROI in a bottom–up manner [16] to capture both the spectral and spatial features of objects, predominantly depending on the characteristics present in the image. The latter is influenced by downstream tasks and falls under a top–down form [46], so this particular approach requires thorough analysis in conjunction with the specific tasks. For instance, a saliency-based HSI detection task needs to be given the location information of the target in the scene through a saliency map, while a saliency-based HSI segmentation task calls for depicting the edge details of the foreground to improve the segment accuracy. However, these methods mainly rely on natural image saliency, without giving a clear definition about HSI saliency in detail.

Therefore, it is necessary to develop a distinctive method for effectively analyzing and processing hyperspectral data. More importantly, the saliency analysis for HSIs demands an in-depth consideration of several critical aspects.

1) The essence of HSA lies in quantifying the visually prominent information across diverse regions, encompassing not only the spatial dimension but also the spectral dimension.
2) HSA should have the ability to assign close saliency values to the regions exhibiting similar spectral attributes and spatial labels.
3) The main purpose of HSA is to highlight the ROIs while eliminating excess noise interference in both spatial and spectral dimensions.
4) The output of HSA should have the same size and dimension as the input HSI.

Guided by the above considerations, we conceived a novel self-supervised saliency analysis (S3A) framework, especially for HSIs. As unveiled in Fig. 1(b), the saliency analysis results on the IP dataset exhibit prominent highlights in regions with land-cover labels and suppress background noise. Moreover, the spectral saliency maps in Fig. 1(f) and (g) reveal varying degrees of salience across different spectral bands, emphasizing the capability to highlight bands of interest while suppressing discordant clutter within the spectral dimension.

Building on the above analysis, we reinterpret the concept of HSI saliency by developing several sound principles, which are presented in Section III-A. Accordingly, our S3A framework is devised to locate the desired regions with less information redundancy and more representation stability. Please note that we regard the HSA as a discriminative task, but not a strict ill-posed problem [47] with various visual probability distributions. It is well-known that the general classification network can output the highest probability for a certain category through softmax layer [38]. To some extent, the variance in the overall distribution of these probability scores serves as an indicator of classification accuracy, subtly reflecting both the image quality and its information content.

Hence, we propose the classification-based patch spectral discrimination network (CPSD-Net) to identify HSI saliency by predicting the score of each patchwise band using the classification network. This allows us to represent the overall saliency within the HSI cube, emphasizing the most distinctive pixel locations and spectral points. Besides that, the design can break away from the strong dependence on specific network structures, such as encoder–decoder architectures, and multifarious loss functions, providing a more robust and effective route for the HSA task. In addition, to provide relatively reliable classification training labels for CPSD-Net, we automatically generate pseudo-labels by utilizing the spectral features of each pixel. Based on this, we design a representative patch sampling (RPS) module, which simply uses unsupervised clustering to automatically choose stronger representative samples without any supervised signal. Also, we develop a patch-to-global spatial diffusion (P2G-SD) module to help homogeneous regions [39] obtain similar saliency values in order to quickly output global hyperspectral saliency.

The main contributions are summarized as follows.

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1) We reinterpret the concept of HSI saliency by developing several sound principles as theoretical support, as detailed in Section III-A. To the best of authors' knowledge, this is the first work to provide a clear and specific definition, especially for HSI saliency.
2) We propose a novel HSA framework S3A integrated with the RPS, CPSD-Net, and P2G-SD modules in a self-supervised manner, which combines the advantages of supervised performance control and unsupervised method generalization. For HSI, S3A generates a stable 3-D saliency cube that accurately highlights important information in the hyperspectral data while effectively suppressing noise.
3) We successfully apply the 3-D saliency cube of HSI to multiple downstream tasks, including BS and HSI classification. Finally, we carry out extensive experiments on various HSI datasets. The quantitative and qualitative results not only demonstrate the reasonability of S3A, but also prove that our S3A framework provides a feasible route for the analysis and application of hyperspectral data.

The rest of this article is organized as follows. Section II reviews related work. Section III outlines the details of the proposed methodology. The results and effectiveness analysis of our method are shown in Section IV. The conclusion is summarized in Section V.

II. Related Work

A. Natural Image Saliency

Visual saliency analysis has been proven effective in natural image processing [29], [30], [31], with most of these methods relying on foundational spatial visual cues, such as contrast, structure, and texture in scenes. Specifically, Li et al. [18] developed principles for designing the saliency detector in the frequency domain, adopting a template-based contrast computation approach. Shi et al. [48] proposed a region-based multilayer hierarchical inference algorithm to tackle the adverse effects of images with small-scale high-contrast patterns in either the background or foreground. Fang et al. [20] proposed to estimate image saliency by learning a set of discriminative subspaces on random contrasts to pop out targets and suppress distractors. The above methods aim to emphasize the discriminative representation between the target and the background by using certain visual attributes, but these features are mostly based on the spatial features of the image itself and lack the semantics of the target, resulting in low robustness of the algorithm.

In recent years, convolutional neural networks (CNNs) have been utilized to explore image saliency and have achieved great success. Recasens et al. [21] introduced a saliency-based distortion layer to improve the ability to perform spatial sampling from the original input data, and the design effectively combines image saliency with the architecture of CNNs, thereby improving the performance of visual tasks. Ji et al. [22] put forth a saliency-guided attention network (SAN), which builds an asymmetrical connection between vision and language based on saliency to learn a fine-grained cross-modality correlation. Zhang et al. [23] presented a saliency-driven unequal single-image reconstruction network, which adopts a gradient-based saliency analysis method and a saliency-driven gate conditional generative adversarial network to improve the performance of remote sensing image super-resolution effectively. Liu et al. [24] designed a two-stream capsule network (CapsNet) to describe how to use the property of part-object relations endowed to fundamentally hinge on relational inference for visual saliency detection, and it effectively addressed the issue of incomplete segmentation of salient objects due to inferring the saliency of each image part separately. Tang et al. [25] proposed a novel concept of structural intersaliency relations and solved the co-salient object detection problem in image groups using a deep reinforcement learning framework.

These deep-learning-based methods aim to capture strong deep discriminative features of targets and backgrounds through CNNs, but they are mostly limited to the spatial dimension and are not well-suited when processing high-dimensional image data [49]. In addition, natural image visual saliency is mostly task-driven in a top–down manner, making it difficult to achieve all-round saliency measurement. For example, the detection task needs to highlight the location of the target, while the segmentation task is more sensitive to the edge of the target. Therefore, it is imperative to generalize the concept of visual saliency and promote it to higher dimensions.

B. HSI Saliency

Due to the brilliant assistance of image saliency in visual tasks, the concept of image saliency has been applied to hyperspectral data processing tasks [40], [41], [42]. The existing works can be approximately divided into two classes, feature-compared HSI saliency and task-driven HSI saliency. For instance, Le Moan et al. [43] defined the concept of spectral saliency as the degree to which a group of pixels stand out in an image and in terms of reflectance feature, rather than in terms of chromaticity attributes, as in traditional saliency studies. The proposed model can output saliency maps through a series of customized FE, not only depicting which objects are prominent, but also depicting the wavelength distribution range. However, this method employs manual features, resulting in poor robustness of the model. Shen et al. [44] proposed a purity-aware saliency method, which calculates purity feature through abundance fraction vector norm, but this method relies solely on spatial information, neglecting the spectral dimension. Su et al. [45] presented a BS method based on saliency band and scale selection (SBSS), which defined the expected band with certain properties as saliency band. Moreover, to demonstrate the effectiveness of SBSS, three key prerequisite assumptions are proposed: first, exploring the intrinsic properties of hyperspectral data requires combining spectral and spatial information. Second, the saliency band can approximately represent all of the information of HSI. Third, there is an inherent connection between saliency, scale, and image description. Shen et al. [41] proposed a matrix decomposition method with saliency priors to handle the problem of anomaly detection, which uses saliency prior knowledge to measure the potential of each pixel to be an anomaly.

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However, most of the current work related to saliency is driven by visual tasks, while specific HSI saliency research is relatively scarce. Although the above methods claim to adopt visual saliency, they still follow the methods of natural image saliency, without considering the characteristics of HSIs and the task itself. In this article, we reinterpret the concept of HSI saliency based on the inherent characteristics of HSI and develop several reasonable principles. At the same time, we design a self-supervised HSA framework and successfully apply the saliency analysis results to hyperspectral downstream tasks including BS and HSI classification.

C. Hyperspectral Feature Analysis

Hyperspectral feature analysis plays a crucial role in high-dimensional spectral data processing by extracting essential information while mitigating redundancy [6]. To achieve this, three key techniques such as BS [33], [34], [35], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], band sampling [36], [37], and FE [38], [39], [62], [63], [64] are commonly employed to provide more representative and discriminative features for downstream tasks.

BS aims to identify the representative bands while preserving their physical interpretability. Existing approaches are primarily guided by physical priors and statistical metrics, such as uniform BS (UBS) [33], information entropy-based band ranking [35], [50], [51], [57], clustering-driven representative band screening [52], [58], [59], [60], and greedy search for optimal band subset determination [65], [66]. For instance, Li et al. [33] proposed a uniform band interval selection method that partitions the spectral space into evenly distributed intervals, selecting representative bands to ensure spectral integrity. Wang et al. [57] introduced a self-supervised deep multilevel representation learning framework with entropy-based clustering to enhance spectral discriminability. Tang et al. [58] developed a density peak-based clustering algorithm, leveraging local density and distance metrics for robust BS. Yu et al. [66] employed an orthogonal subspace projection approach to estimate band-specific virtual dimensionality, optimizing informative BS.

In contrast, band sampling [36], [37] provides an alternative strategy by selecting a subset of spectral bands while maintaining the overall spectral distribution. For example, Chang [36] introduced a statistical band sampling framework, which systematically reduces spectral redundancy while maintaining key spectral characteristics. Chang et al. [37] further extended this concept with an anomaly-space-based band sampling method, designed to enhance hyperspectral anomaly detection by maximizing the retention of anomaly-relevant spectral information. In addition, although FE techniques [62], [63], [64] can improve feature representation, they often compromise the physical integrity of hyperspectral data. Consequently, BS and band sampling are widely used for dimensionality reduction while retaining spectral interpretability for downstream analysis.

III. Methodology

In this section, we first reinterpret the concept of HSI saliency using several robust principles, then elaborate on the details of our S3A framework, and finally, we describe how to detect the hyperspectral saliency in both spectral and spatial domains for the specific applications.

A. Principles of HSI Saliency

Hyperspectral data are typically different from RGB images due to its rich, high-resolution continuous spectrum information, which provides reliable visual clues for accurate material classification and data mining. Consequently, when analyzing HSI, it is necessary to consider its properties, including the uniqueness and redundancy of material spectra, as well as the homogeneity and differences of spatial features, rather than simply adopting natural image analysis methods. To address this, we introduce the concept of HSI saliency for the first time by effectively merging the distinctive attributes of HSI with the saliency criteria observed in natural images [2], [67], [68]. Building on the in-depth consideration of several critical aspects discussed in Section I, the integration is guided by four fundamental principles, which are outlined as follows.

1) *Highlight the Most Discriminative Spectral Bands:* Spectral information reflects the classes of land-cover objects and can be used as the basis for identifying the type and composition of materials. Therefore, it is necessary to emphasize the obvious discriminated spectrum, which is highly salient, to perceive most or even all land-cover classes contained in HSI.
2) *Equally Emphasize the Homogeneous Spatial Regions:* Homogeneous regions [39] in HSI contain the same spectral characteristics, which means that these regions include similar saliency. As long as the saliency of a part of the region is calculated, its homogeneous regions can be approximately obtained through quick dissemination.
3) *Disregard the Noisy Bands:* The noisy bands should be suppressed, as they lack the capability to effectively characterize the properties of land-cover objects, so the saliency should be kept extremely low to avoid potential interference caused by noise.
4) *Output Full-Resolution Saliency Cube to HSI Size:* The saliency output of HSA should be a full-resolution cube with the same dimensions as the input HSI, ensuring compatibility and facilitating downstream application tasks. Furthermore, by preserving the original data structure, we can seamlessly integrate the saliency analysis results into existing HSI processing pipelines, thereby enhancing the overall performance and accuracy of HSI interpretation.

In essence, the HSI saliency analysis conducted through these four pivotal principles effectively reveals characteristics embedded within hyperspectral data. By accentuating vital spectral attributes and spatial properties, HSI saliency can highlight the most conspicuous features or targets within the entire image cube space.

B. S3A Framework

In this section, we describe the components of our S3A framework in detail, as illustrated in Fig. 2. The framework mainly consists of three parts.

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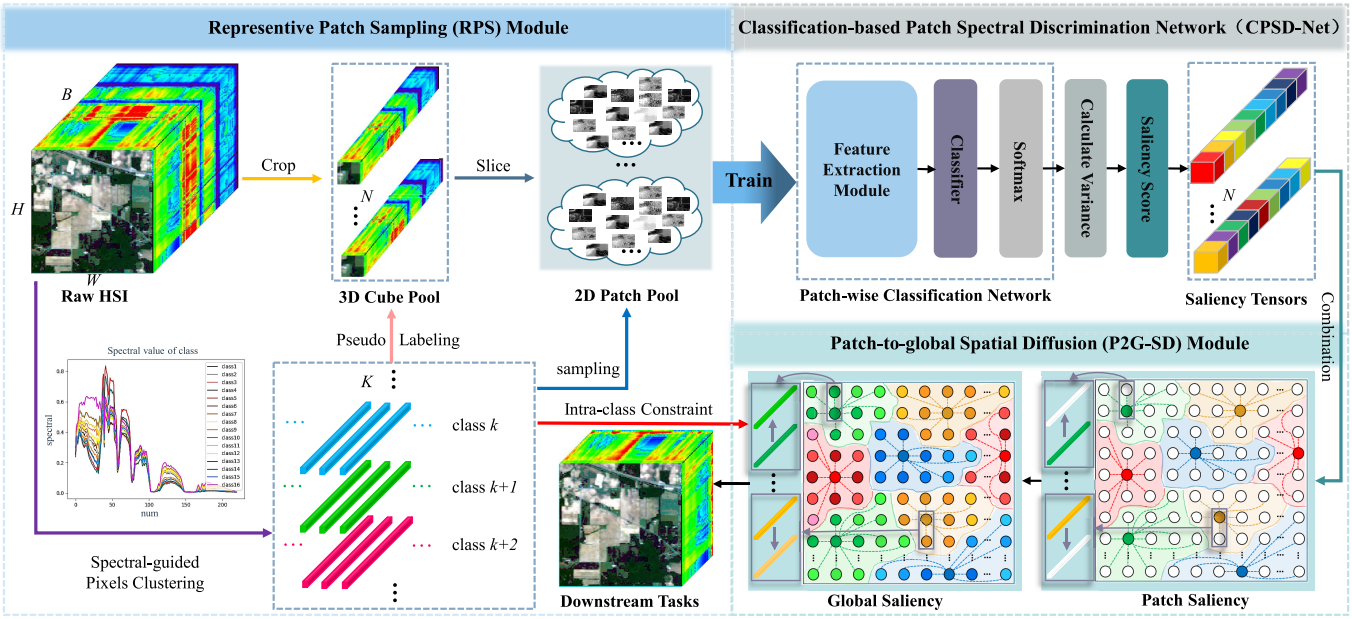


Fig. 2. Overview of our proposed S3A framework, which is composed of three main components, including RPS, CPSD-Net, and P2G-SD modules. Note that the saliency map produced by S3A is consistent with the size of the input HSI data, and then, the HSA cube can be applied to downstream tasks.

1) The RPS module is designed to perform data preprocessing on the original HSI, including cropping and slicing, to pick up some stronger representative samples for self-supervised training to address the challenge of limited labeled data.
2) The CPSD-Net adopts the general image classification model as the backbone to extract features of input patches, which can be used to evaluate the HSI saliency from patchwise spectral discrimination.
3) The P2G-SD module can diffuse the saliency scores to global HSI pixels based on the multidimensional nonlinear filtering and characteristics of homogeneous regions.

Moreover, we integrate these three parts to complete HSA efficiently in a collaborative manner, which is illustrated in Algorithm 1. The details are presented as follows.

1) RPS Module: Assuming the spatial size and band dimension of the original HSI are $H \times W$ and B, the input data can be represented as $X \in \mathbb{R}^{H\times W\times B}$. We define each pixel as a spectral vector $\mathbf{px}_{i,j} = [px_{i,j,1}, px_{i,j,2}, \ldots, px_{i,j,B}] \in X$, where $i \in [1, W]$ and $j \in [1, H]$ denote the pixel spatial location. Based on the material characteristic indicated by the pixel spectral curve, each pixel in the entire HSI can be categorized into one of the C categories, represented by $Y = (y_1, y_2, \ldots, y_C)$.

However, the lack of annotation for hyperspectral data is a common problem, so researchers prefer to automatically obtain the category annotation of each pixel based on its data characteristics. To assign a pseudo-label to each pixel in an unsupervised manner, we employ a spectral-guided pixel K-means [69] clustering method based on Euclidean distance. Regarding each pixel of the HSI as an independent element, its spectral curve is represented as a spectral vector $\mathbf{px}_{i,j} \in \mathbb{R}^{1\times B}$. Between two pixel spectral vectors $\mathbf{px}_{i,j}$ and $\mathbf{px}_{m,n}$ with B bands, their Euclidean distance is calculated via the following function:

$$\mathrm{dist}(\mathbf{px}_{i,j}, \mathbf{px}_{m,n}) = \sqrt{\sum_{b=1}^{B}(px_{i,j,b} - px_{m,n,b})^2}. \quad (1)$$

Algorithm 1 Our Proposed Algorithm for HSA

Preparation: Model construction and initialization
Build classification-based patch spectral discrimination network;
Set the essential hyperparameters Θ_{hp} of the model;
Random initialize the trainable parameters Θ_{tp}.
Input: Input a HSI data $X \in \mathbb{R}^{H\times W\times B}$
Crop X into $\mathbf{x}_{i,j}$ to build a 3D cube pool;
Slice $\mathbf{x}_{i,j}$ into $x_{i,j,b}$ to construct several 2D patch pools;
Regard the patch sample as: $x_{i,j,b} = x_t$;
Cluster pixels and generate pseudo labels $\hat{Y}$;
Format the data sample as $(x_t, \hat{y}_t)$;
Regard each 2D patch pool as a training mini batch B.
Begin
while $iter <= epoch$ **do**
  for x_t in B **do**
    Extract feature of x_t: $\mathbf{o}_t \leftarrow f(\psi x_t + \beta)$
    Obtain prediction vector: $\mathbf{p}_t \leftarrow softmax(\mathbf{o}_t)$
    Calculate variance s: $s_{i,j} \leftarrow s_t \leftarrow s(\mathbf{p}_t)$
    Normalize weight: $\mathbf{S_{i,j}} \leftarrow Normalize(s_{i,j})$
  end for
end while
Diffuse patch sample weights to global pixel according to P2G-SD module;
End
Output: The HSI global 3D saliency cube $S \in \mathbb{R}^{H\times W\times B}$.

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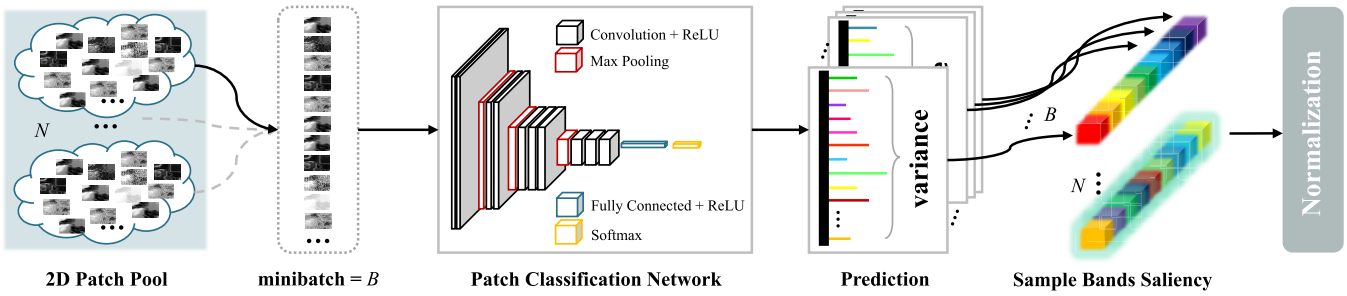


Fig. 3. Detailed flowchart of CPSD-Net. It should be noted that the total number of 2-D patch pools is N, and each 2-D patch pool is composed of B samples. Therefore, the output of this layer is N weight vectors with dimension B. In addition, here, we only take the VGG network as a backbone example of the patch classification network.

Until all pixels are ultimately aggregated into K clusters, which is determined based on domain knowledge such as approximate land-cover class counts, the cluster centers can be expressed by $\hat{Y} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_K)$.

When assessing saliency in HSIs, it is crucial to consider both spatial and spectral information. Therefore, we crop the entire HSI along the spatial and spectral dimensions during data preprocessing to construct corresponding sample pools. Spatially, we crop a neighboring region of size $P \times P$ with default value $P = 27$ to represent the center pixel (i, j) in the original HSI, building a 3-D patch subcube $\mathbf{x}_{i,j} \in \mathbb{R}^{P \times P \times B}$. Then, the input HSI can be represented as a total of $H \times W$ 3-D patch subcubes. Spectrally, each 3-D patch subcube is sliced along the spectral dimension to obtain multiple 2-D spectralwise patch samples, and multiple patch samples $x_{i,j,b} \in \mathbb{R}^{P \times P}$, where $b = 1, 2, \ldots, B$, are obtained to construct distinct 2-D patch pools. These 2-D patches form distinct patch pools for each spectral band, allowing us to analyze the discriminative power of each band separately. After the above process, we establish the necessary sample pools and assign appropriate pseudo-labels to each sample for subsequent self-supervised HSA in CPSD-Net.

2) Classification-Based Patch Spectral Discrimination Network: To explicitly estimate the discriminability of different spectral bands for each pixel, we design CPSD-Net to obtain a saliency score for each band patch via classification prediction probabilities. In Fig. 3, we show the detailed process of generating pixel band saliency weights through CPSD-Net, where the backbone of CPSD-Net can be adapted to various image classification networks.

The CPSD-Net consists of two components, an FE module and a classifier. The FE module mainly extracts representative robust features through manual design or neural networks, and the classifier is used to predict the class probability, which assigns the class with the maximum probability to the pixel. In addition, the softmax is usually attached to the last layer of the classifier. The entire image classification process can be described as

$$\mathbf{o_t} = f(\psi x_t + \beta) \tag{2}$$

$$\mathbf{p}_t = \text{softmax}(\mathbf{o_t}) \tag{3}$$

where x_t represents the tth training sample, which is taken from the constructed 2-D sample pools by $\mathbf{x}_{i,j}$, and $t \in [1, B]$. ψ, β, and $f(\cdot)$ are the weights, biases, and fully connected layers, respectively. $\mathbf{o_t}$ is the feature vector after the FE module, and $\mathbf{p}_t$ is the class probability vector predicted by the classifier. The predicted result is the class corresponding to the maximum probability, which can be expressed as

$$\tilde{y}_t = \text{argmax}(\mathbf{p_t}). \tag{4}$$

The cross-entropy loss is adopted to measure the distribution between the predicted and real labels, which is widely used in image classification tasks. The loss function is defined as

$$L = -\frac{1}{N'} \sum_{t=1}^{N'} \sum_{k=1}^{K} 1\{\hat{y}_t = \hat{Y}(k)\} \log\left(p_t^k\right) \tag{5}$$

where $1\{\hat{y}_t = \hat{Y}(k)\}$ is an indicator function that equals 1 when the predicted label $\hat{y}_t$ matches the kth class label, and 0 otherwise. N' represents the total number of samples in the training process.

In the RPS module, the whole HSI pixels can be clustered into K clusters. Thus, the class probability vector $\mathbf{p}_t$ obtained after softmax should contain K dimensions, which can be described as $\mathbf{p_t} = [p_t^1, p_t^2, \ldots, p_t^K]$. After the above classification process, each band patch can get a probability distribution of its class. It is worth noting that the probability distribution can directly reflect the discriminability of the patch sample. In other words, the performance in the classification task can be analogized to the amount of category properties it contains. Therefore, we describe the probability distribution through variance statistic, which can be formulated as

$$s_t = \sqrt{\frac{\sum_{k=1}^{K} \left(p_t^k - \text{average}(\mathbf{p_t})\right)^2}{K - 1}} \tag{6}$$

where p_t^k represents the prediction probability corresponding to the kth class. Assuming that the tth training sample is taken from the bth band patch of pixel $\mathbf{x}_{i,j}$, then $s_{i,j,b} = s_t$.

During the training process, we take the 2-D patch pool constructed from each HSI pixel as an input mini-batch. After each iteration, the variance statistical characteristics of the discriminative prediction probabilities across all patch images at each pixel can be obtained. The larger the variance, the greater the maximum probability of predicting a certain category, which means the sample is easier to identify. Meanwhile, the sample contains more category discriminative information, which is more salient than other samples. And it should be assigned a larger weight to highlight the most discriminative spectral

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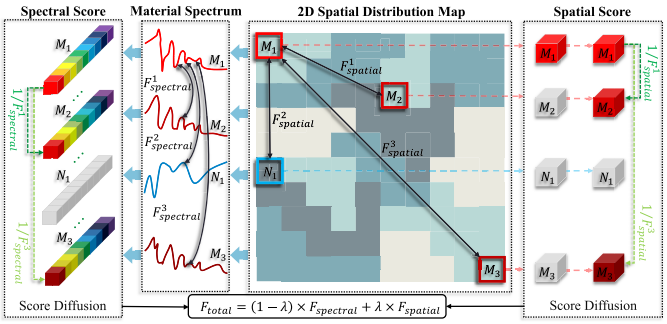


Fig. 4. Detailed flowchart of P2G-SD. It should be noted that M_1, M_2, and M_3 represent the spatial positions holding the same land-cover object, while N_1 is another different land-cover objects. The diffusion score is related to spectral similarity and spatial distance, which is weighted by the spatial score and spectral score with a tradeoff parameter λ.

bands during saliency analysis. Furthermore, we employ a min–max normalization method to normalize the generated weights as band patch saliency, to ensure that the weight vector of each sample is distributed in the [0, 1] scale range

$$\mathbf{S_{i,j}} = \text{Normalize}\left(\left[s_{i,j,1}, s_{i,j,2}, \ldots, s_{i,j,B}\right]\right). \tag{7}$$

3) P2G-SD Module: The saliency of each input pixel can be obtained through the CPSD-Net. However, when we regard all band patch samples of each pixel as a mini-batch, the CPSD-Net can only calculate one pixel after completing the forward process in each iteration. If all HSI pixels participate in the training process, it will undoubtedly bring a huge amount of calculation. Considering that homogeneous regions in HSI contain the same spectral characteristics, it means that these regions include similar salience. As long as the saliency of a part of the region is calculated, its homogeneous regions can be approximately obtained through quick dissemination.

However, due to the phenomenon of the "same substance with different spectrums" caused by the differences in illumination and radiation in diverse spatial regions, we also need to consider the spectral characteristic distribution of the image patches with the same class located in different spatial areas. Therefore, we design a novel multidimensional nonlinear filtering method that can quickly diffuse saliency scores to global HSI pixels.

Fig. 4 shows the entire process of diffusing spectral-wise pixel salient scores to all points in the entire HSI through P2G-SD. During the diffusion process, the scores are only propagated to homogeneous regions based on both spatial distance F_{spatial} and spectral similarity F_{spectral}. Here, M_1, M_2, and M_3 represent homogeneous regions located at different spatial positions but containing the same land-cover object, while N_1 is another different land-cover object. Since there is no strong correlation between different categories of land-cover objects in nonhomogeneous regions, the saliency score of N_1 is not related to M_1, M_2, and M_3. Given M_1 as the reference point, the spatial distance F^1_{spatial} of M_2 is lower than F^3_{spatial} of M_3, so the spatial diffusion score of M_2 should be greater than the spatial diffusion score of M_3, when diffusing from M_1. At the same time, considering that the spectral distance F^1_{spectral} of M_2 is less than F^3_{spectral} of M_3, the spectral diffusion score from M_1 to M_2 should be higher than M_3. Finally, the multidimensional nonlinear filtering method is calculated from the spatial and spectral perspectives, and it is designed in the form of a dissemination function.

Assuming the saliency score vector of pixel $\mathbf{px}_{i,j}$ is $\mathbf{S}_{i,j}$, if the pixels $\mathbf{px}_{i,j}$ and $\mathbf{px}_{m,n}$ belong to the same category, then the saliency score vector $\mathbf{S}_{m,n}$ of pixel $\mathbf{px}_{m,n}$ can be obtained by $\mathbf{px}_{i,j}$, and the dissemination process can be defined as

$$F_{\text{spatial}} = \sqrt{(i-m)^2 + (j-n)^2} \tag{8}$$

$$F_{\text{spectral}} = \sqrt{\sum_{b=1}^{B} \left(px_{i,j,b} - px_{m,n,b}\right)^2} \tag{9}$$

$$F_{\text{total}} = (1-\lambda) \times F_{\text{spectral}} + \lambda \times F_{\text{spatial}} \tag{10}$$

$$\mathbf{S}_{m,n} = \frac{1}{F_{\text{total}}} \times \mathbf{S}_{i,j} \tag{11}$$

where F_{spatial}, F_{spectral}, and F_{total} represent the spatial distance, spectral difference, and total dissemination measurement for different pixels, respectively. λ is a tradeoff parameter used to adjust spatial and spectral importance. As a rule of thumb, the λ value is set to 0.5. $\mathbf{S}_{m,n}$ is inversely proportional to the value of F_{total}. Finally, we collaboratively integrate the above RPS, CPSD-Net, and P2G-SD into our S3A framework to obtain a 3-D saliency cube $S \in \mathbb{R}^{H \times W \times B}$.

C. Analysis of HSI Saliency

HSA should be conducted within specific application scenarios, as different tasks come with distinct requirements and criteria for saliency assessment. For instance, in BS task, spectral saliency is generally more important, while for tasks involving detection and classification, spatial saliency garners greater attention. As a result, our S3A approach involves a comprehensive examination of HSI saliency, considering both spectral and spatial aspects in practical tasks.

1) Look for Saliency in Spectral Domain: For the obtained 3-D HSA cube, when observing from the perspective of a single pixel, the spectral saliency can be expressed as a vector, similar to a spectral curve. Moreover, it can also be observed from a certain interested spatial region of the HSI, and we can get a saliency curve in the spectral domain when integrating the spatial saliency values, such as pixels from the same cluster. It is worth noting that pixels with the same class label tend to have similar spectral saliency curves, which is consistent with the HSA principles we explained. If the interested region is extended to the entire HSI, the spectral saliency is equivalent to the band saliency, and then, the spectral saliency curves quantify the discriminative ability of each spectral band, providing valuable assistance for the BS task. Hence, we introduce HSI saliency into BS to assist in measuring the importance of spectral bands. By fully leveraging spectral saliency, a subset of spectral bands can be selected to effectively capture the essence of the entire HSI dataset. The spectral saliency is formally described as follows:

$$S^{(\Upsilon)}_{\text{spe}} = \frac{1}{g_\Upsilon} \sum_{(i,j) \in \Upsilon} \mathbf{S}_{i,j} \tag{12}$$

where $S^{(\Upsilon)}_{\text{spe}} \in \mathbb{R}^{1 \times B}$ represents the average spectral saliency in interested spatial region Υ with pixel number g_Υ.

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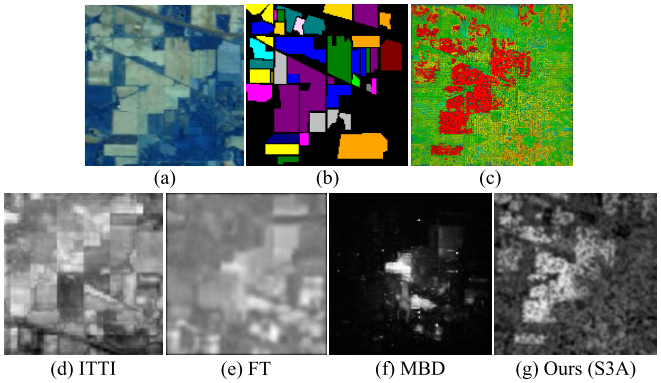


Fig. 5. Visualization result of saliency map in the spatial domain on IP dataset. (a) False-color image. (b) Land-cover category labels. (c) Spatial-dimension saliency produced by our S3A. (d)–(g) Saliency results produced by different RGB image saliency methods from false-color images and our S3A smooth-filtered spatial saliency results.

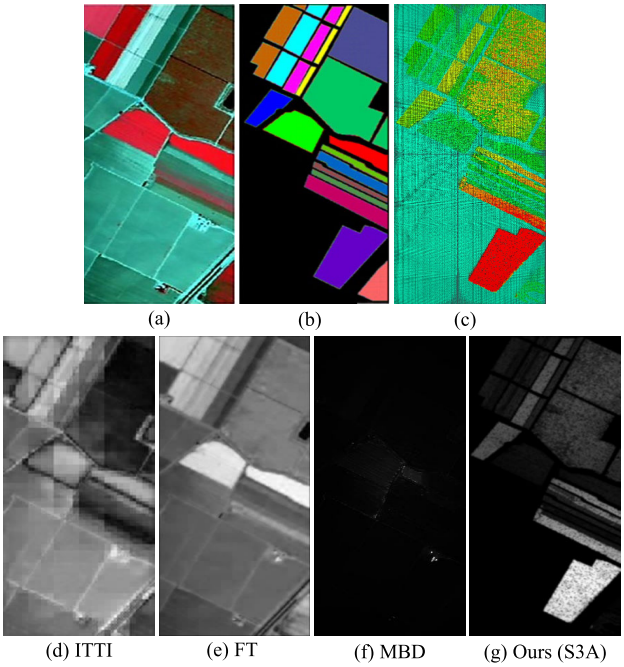


Fig. 6. Visualization result of saliency map in the spatial domain on SA dataset. (a) False-color image. (b) Land-cover category labels. (c) Spatial-dimension saliency produced by our S3A. (d)–(g) Saliency results produced by different RGB image saliency methods from false-color images and our S3A smooth-filtered spatial saliency results.

2) Look for Saliency in Spatial Domain: The spatial saliency [57] is similar to the saliency of natural images, where the saliency at each pixel position can be obtained by averaging the spectral saliency curve of that pixel. Additionally, spatial saliency can be analyzed within a specific band subset of interest in the HSI, resulting in a saliency map in the spatial domain through the integration of spectral saliency values. The spatial saliency reflects the importance of each pixel position, which has great guiding significance for the precise recognition and classification of targets. The spatial saliency can be described as follows:

$$S_b \overset{\text{def}}{=} \left\{ S_{i,j,b} : \forall (i,j) \in \mathbb{R}^{H \times W} \right\} \tag{13}$$

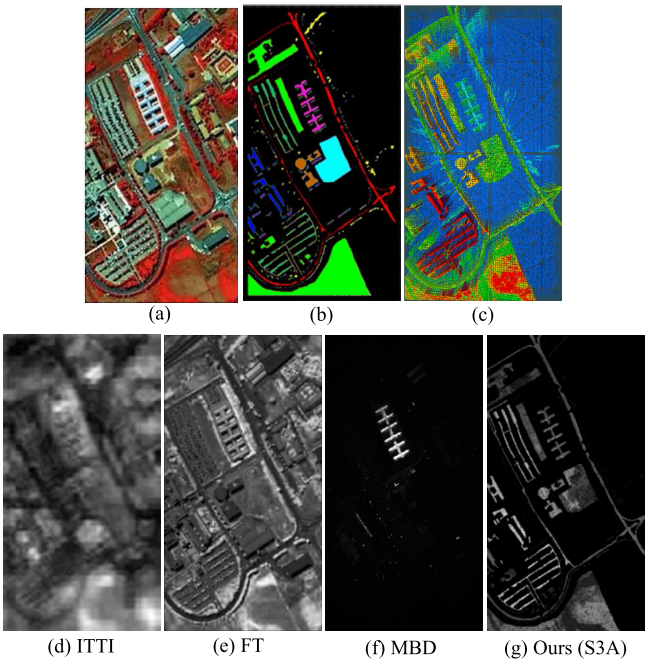


Fig. 7. Visualization result of saliency map in the spatial domain on PU dataset. (a) False-color image. (b) Land-cover category labels. (c) Spatial-dimension saliency produced by our S3A. (d)–(g) Saliency results produced by different RGB image saliency methods from false-color images and our S3A smooth-filtered spatial saliency results.

$$S_{\text{spa}}^{(B')} = \frac{1}{g_{B'}} \sum_{b \in B'} S_b \tag{14}$$

where S_b is the saliency in bth spectral band, and $S_{\text{spa}}^{(B')} \in \mathbb{R}^{H \times W \times g_{B'}}$ represents the spatial saliency of the whole HSI region with $g_{B'}$ interested spectral bands B'.

IV. Experimental Results

In this section, to verify the feasibility and effectiveness of our S3A on different HSI datasets, we compare the performance with other algorithms on different downstream tasks including BS and HSI classification, and conduct ablation experiments to demonstrate the rationality of each part of the framework.

A. Experimental Dataset

We conduct all experiments on three public HSI benchmark datasets, which are described in detail as follows.

1) Indian Pines: IP dataset [70] is imaged by NASA/JPL's Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in June 1992, and the scene covers an agricultural area in northwestern Indiana's India pine test site, USA. The dataset is intercepted with a spatial size of 145 × 145 for the HSI classification application, and the spatial resolution is 20 m/pixel. The spectral range spans from 0.4 to 2.5 μm with a resolution of approximately 10 nm. It totally consists of 220 consecutive bands, but part of them are noise bands because they cannot be reflected by water, including the 104th–108th, 150th–163rd, and 220th bands. The overall IP dataset contains 16 different land-cover classes.

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2) Salinas: Salinas (SA) dataset [71] is also taken by AVIRIS imaging spectrometer, which is an image of SA Valley in California, USA. Different from IP dataset, its spatial resolution reaches 3.7 m. The original data consist of 224 bands, but part of them are noise bands because they cannot be reflected by water, including the 108th–112th, 154th–167th, and 224th bands. The spatial size of SA dataset is 512 × 217 and is totally divided into 16 different land-cover classes.

3) Pavia University: Pavia University (PU) dataset [72] was acquired by the Germany Reflective Optics System Imaging Spectrometer (ROSIS) sensor, which covers the Italian city of Pavia in 2003. It has 103 spectral wavelengths in total ranging from 0.43 to 0.86 μm. The spatial size of PU dataset is 610 × 340 with 1.3-m/pixel spatial resolution. The whole PU dataset contains nine different land-cover classes.

B. Visualization of HSI Saliency

HSA task requires joint spectral–spatial dimensional information to identify salient regions. Therefore, the generated HSA cube should be reflected in both the spectral and spatial domains, and exist in different spatial distribution positions and spectral curve subsets. To observe the HSI saliency intuitively, we visualize the HSI saliency from spatial and spectral views, respectively.

1) Saliency in Spatial View: We visualize the spatial saliency results on different HSI datasets, as shown in Figs. 5–7. We performed Gaussian smoothing filtering on our spatial saliency map. From the spatial saliency results, it can be observed that compared with other natural image saliency methods, our S3A can emphasize more foreground useful information and suppress complicated background, and it can highlight prominent positions in the target area of the image. Furthermore, the brighter the intensity in the saliency map, the easier it is to distinguish the category. Finally, the spatial size of the entire saliency map is consistent with the spatial size of the original HSI, and the homogeneous regions exhibit similar intensity patterns, consistent with the second and fourth principles of our HSA framework.

2) Saliency in Spectral View: We also visualize the results of spectral saliency, as shown in Fig. 8. From the spectral saliency map of the entire HSI, it can be observed that the spectral saliency of each band is different. Our S3A can highlight the discriminative spectral bands and suppress useless noise bands, which is in line with the first and third principles of our proposed for HSA.

In addition, to illustrate the differences in spectral saliency across classes, we visualize the normalized average spectral saliency curves of each class and mark the top two most salient bands in each class. It can be found that the most salient bands corresponding to each class are generally different, and the trajectory and distribution ranges are quite different. To demonstrate the characteristic of spectral saliency, we multiply the spectral saliency values with the average spectral curve of each class and make a meticulous comparison of the average spectral curves of each class before and after the saliency enhancement. As shown in Fig. 8, it can be found that the original average spectral curves of each class exhibit a greater degree of resemblance, while the saliency-enhanced spectral curves of each class become notably more distinguishable, indicating that our saliency can enhance the spectral discrimination ability of different classes.

C. Downstream Application of HSI Saliency in BS Task

1) Experimental Setup: To validate the adaptability and feasibility of our S3A framework in a fair and objective manner, the representative downstream task BS is chosen as a saliency information carrier. Similar to conventional BS methods, the overall quality of the selected bands is evaluated based on their classification accuracy in terms of three distinct evaluation metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient. Note that for convenience of description, Kappa×100 is uniformly recorded as Kappa in the following content.

With the selected bands, the classification model is trained on a randomly selected 5% of labeled samples from each dataset, while the remaining samples are used for the testing process. We adopt two traditional classifiers k-nearest neighborhood (KNN) and support vector machine (SVM). Note that the number of neighbors is set to 3 in the k-nearest neighbors algorithm, and the SVM is equipped with a radial basis function (RBF) kernel, which is similar to the evaluation strategy adopted in other BS methods. In addition, for our S3A, we first obtain the saliency score of each band of HSI according to 12, when Υ is set to the spatial region of $W \times H$, equal to the spatial resolution of the entire HSI. Then, we sort the bands of the HSI cube according to the averaged saliency score of each band, so as to obtain the expected top-k bands to form the band subset.

In order to conduct a more comprehensive analysis and comparison of the experiments, given the number of selected bands as b, we test the classification performance with the top b' selected subset bands of each BS method, where b' is ranging from 3 to b with about five-band interval. To ensure a reliable result, we implement each comparison method for five times and take the average value as the final result. For our S3A, we randomly selected 20% of data to obtain the saliency score from the CPSD-Net, and the rest data are calculated by integrating the spatial distance and spectral characteristics through the P2G-SD module. Moreover, we adopt the classic AlexNet as the backbone of model and train it for 100 epochs to preserve the best prediction model. The learning rate and batch size are set to 0.02 and 128, respectively.

On the other hand, to quantitatively analyze the information of the selected band subsets, we further calculate the information entropy and mean spectral divergence (MSD) of band subsets. The information entropy can measure the amount of information contained in a band image. The larger the value is, the more information it contains. The HSI band information entropy can be defined as follows:

$$E\left(x_{i,j,b}\right) = -\sum_{\hbar} p(\hbar)\log(p(\hbar)), \ \text{s.t.} \ \sum_{\hbar} p(\hbar) = 1 \quad (15)$$

$$p(\hbar) = \frac{n(\hbar)}{P \times P} \quad (16)$$

where $\hbar$ represents each gray level of histogram bins in HSI patch matric composed of $P \times P$ pixels, and $n(\hbar)$ and $p(\hbar)$

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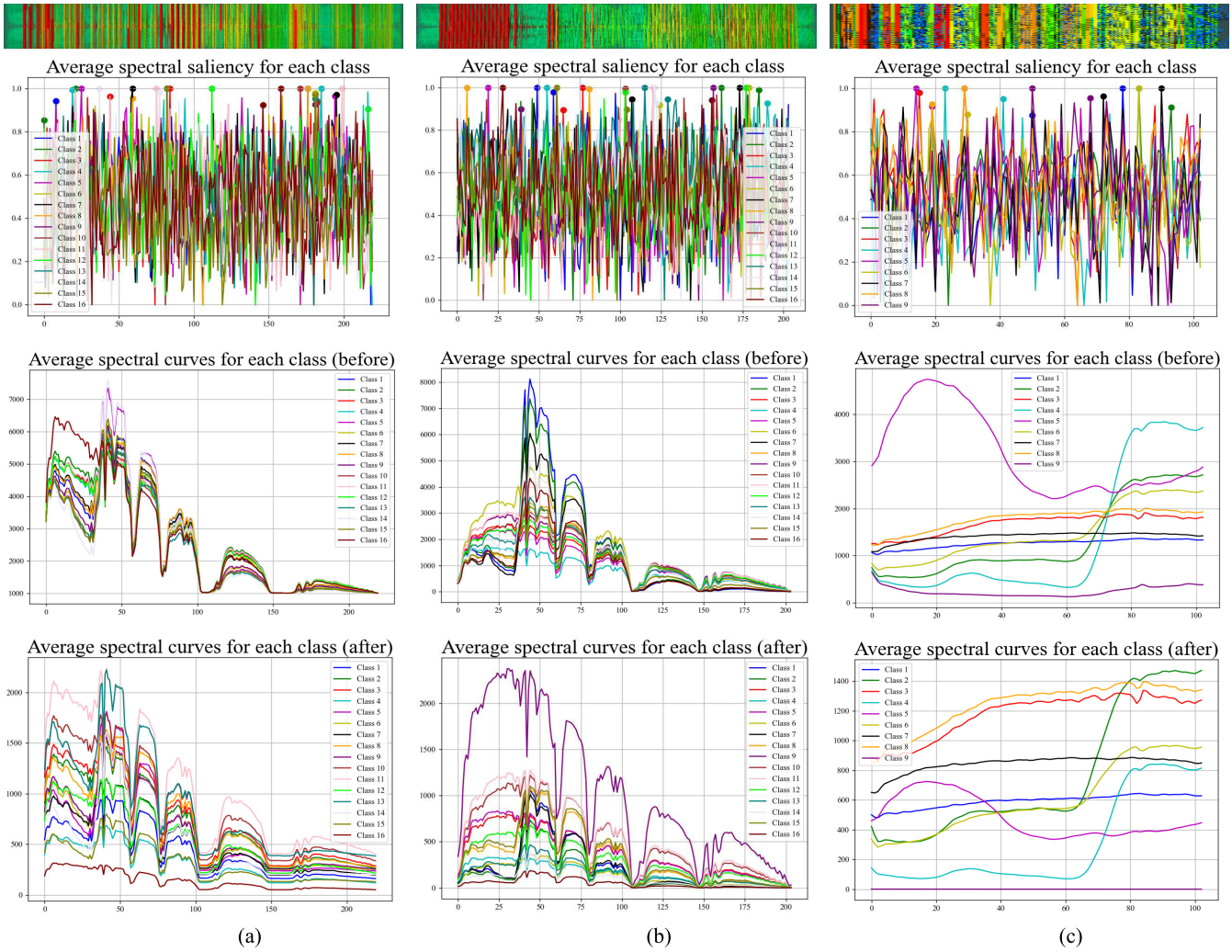


Fig. 8. Visualization result of saliency in the spectral domain on three different HSI datasets. The top-to-bottom rows, respectively, display the spectral saliency map of the entire HSI, the normalized average spectral saliency for each class, the original spectral curves for each class, and the average saliency-enhanced spectral curves for each class. (a) IP dataset. (b) SA dataset. (c) PU dataset.

are the number and probability of $\hbar$ occurs in whole image, respectively. The MSD is a common average measurement index, which can reflect the redundant information contained in selected subsets. Generally, within a certain range, the greater the value is, the less redundant information it contains. It can be expressed as

$$M\left(x_{i,j,p}, x_{i,j,q}\right) = \frac{2}{s(s-1)} \sum_{p=1}^{s} \sum_{q=1}^{s} D_{\mathrm{sKL}}\left(x_{i,j,p} \| x_{i,j,q}\right) \quad (17)$$

where s is the band number of selected subset, and D_{sKL} represents the symmetrical Kullback–Leibler divergence [73]. It is used to measure the distribution difference between $x_{i,j,p}$ and $x_{i,j,q}$ in bidirections, which is expressed as $D_{\mathrm{KL}}(x_{i,j,p} \| x_{i,j,q})$ and $D_{\mathrm{KL}}(x_{i,j,q} \| x_{i,j,p})$, respectively,

$$D_{s\mathrm{KL}}\left(x_{i,j,p} \| x_{i,j,q}\right) = D_{\mathrm{KL}}\left(x_{i,j,p} \| x_{i,j,q}\right) + D_{\mathrm{KL}}\left(x_{i,j,q} \| x_{i,j,p}\right). \quad (18)$$

Note that all the comparison methods and experiments are evaluated in the same environment, which are run on an Intel i9-11900K 3.50-GHz CPU with 64-GB RAM to ensure the fairness of the experiment. In addition, we accelerate all the deep learning-based methods on an NVIDIA GeForce RTX 3090 GPU with 24-GB graphic memory.

2) BS Methods in Comparison: We compared the BS guided by spatial saliency of our S3A with a variety of BS methods, including five popular classical BS methods (UBS [33], ranking-based MVPCA [35], entropy-based IE [50], density-peak-based EFDPC [51], and clustering-based OCF [52]) and four latest unsupervised BS methods (embedding-learning-based RDGSR [53], sparsity-based SRL-SOA [54], self-representation-based S4P [55], and attention-based DARecNet [56]). It is worth noting that we utilize three versions of OCF to participate in the evaluation, namely, NC-OC-MVPCA, OCF-IE, and TRC-OC-FDPC, which are improvements based on MVPCA, IE, and EFDPC, respectively. Moreover, the account of clusters in RDGSR is set to equal the number of dataset classes, which is the same as our S3A, and other parameters are consistent with the original settings. Finally, we implement SRL-SOA with Tensorflow-GPU 2.4.1 version, and the epoch and iteration are set to

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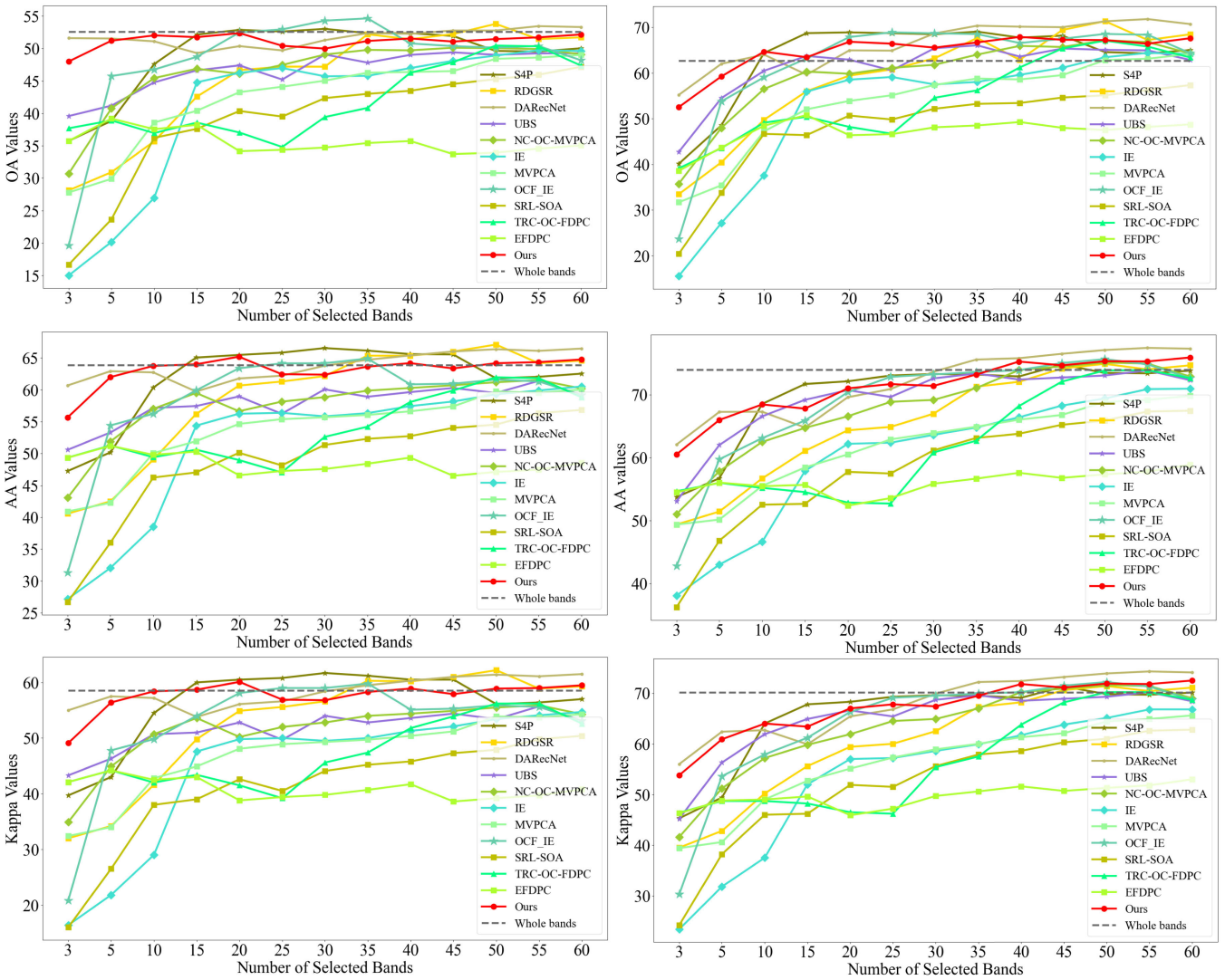


Fig. 9. Performance comparison of different BS methods on the corrected IP dataset. The figure only shows the results of the top 60 bands. (Left column) Results obtained by the KNN classifier. (Right column) Results obtained by the SVM classifier. (First–third rows) OA curves, AA curves, and Kappa curves, respectively.

50 and 10, respectively, which are the same as the official article.

3) Comparison Results on IP Dataset: In order to compare fairly with other methods, we adopt the corrected IP dataset containing a total of 200 bands for the experiment, which is usually used in most BS tasks.

From the results shown in Fig. 9, it can be observed that the classification performance of our S3A method is superior to that of the above algorithms in most band subsets. In particular, our S3A shows obvious advantages when selecting a small number of bands, such as 3, 10, and 15 bands. The general trend of each curve is that the classification performance gradually increases with more selected bands. However, the curves of our S3A can be approximately distributed around the all-band performance, which also reflects the effectiveness of saliency analysis. Furthermore, Fig. 10 shows the BS results in the full HSI bands. As we can see from the results, our S3A method outperforms both the full-band approach and other BS methods in most band subsets. Furthermore, the trend observed in the performance curves indicates that classification accuracy initially improves as the number of selected bands increases, eventually converging toward full-band performance. This can be attributed to the fact that a greater number of selected bands allows for the inclusion of more discriminative features that improve classification. However, when the selected band subset contains noisy or redundant bands, classification performance is adversely affected, leading to potential degradation. This observation is consistent with the well-known "Hughes phenomenon," which states that classification performance initially improves with an increasing number of selected bands but eventually declines beyond a critical point. It should be noted that OCF usually does not work normally when the saliency of all bands is obtained through a training phase, so the results of NC-OC-MVPCA, OCF-IE, and TRC-OC-FDPC will be the same as those of MVPCA, IE, and EFDPC.

In Table I, we show the best 15 bands in the top 60 salient bands and other BS methods. To observe the distribution of these bands more intuitively, we indicate the specific band indices from the original HSI in Fig. 11 and band information entropy. As we know, the entropy values of adjacent bands

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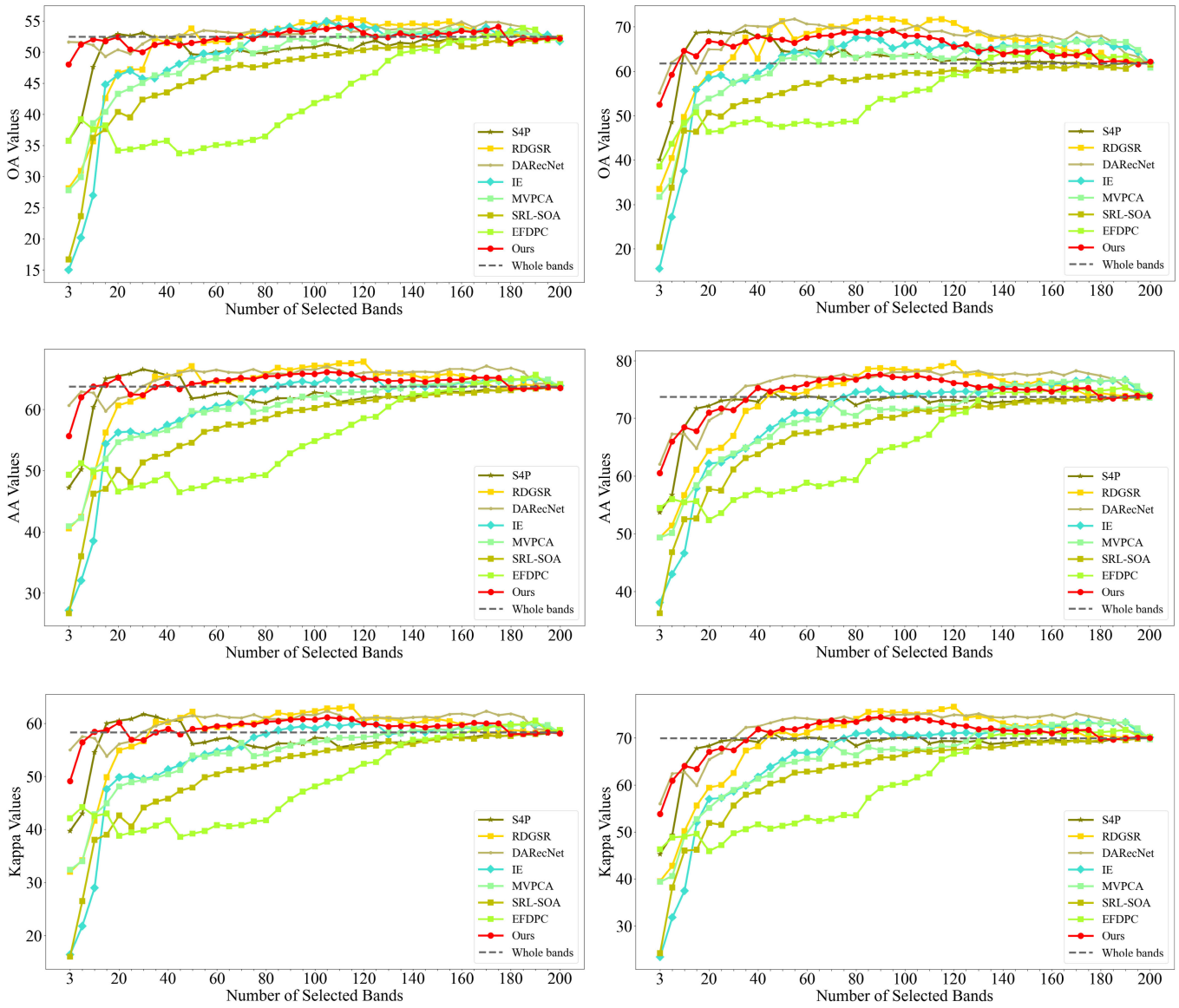


Fig. 10. Performance comparison of different BS methods on the corrected IP dataset. The figure shows the results of 200 bands. (Left column) Results obtained by the KNN classifier. (Right column) Results obtained by the SVM classifier. (First–third rows) OA curves, AA curves, and Kappa curves, respectively.

are similar, and the bad noisy bands are also with extremely low entropy values. From Fig. 11, we only illustrate several typical representatives from all noise bands, such as [104, 105, 144, 145, 198, 199, 200]. The bands selected by our S3A can effectively avoid these bands with low entropy, and effectively reduce the impact of redundant information due to the wide dispersion of the selection bands. In addition, although other BS algorithms can also achieve respectable classification performance, from the distribution point of view, the bands selected by OCF_IE, MVPCA, IE, and NC_OC_MVPCA are mainly distributed in the front of the entire spectrum, and the bands selected by EFDPC and TRC_OC_FDPC are mainly distributed in the back of the full spectrum, while the bands we selected can be approximately evenly distributed in the flat regions of the entire spectrum.

In Fig. 12, we calculate the MSD for band subsets with different band number obtained by the above BS methods. From the results, we can see that the MSD value of our method is better than that of other competitors except for SRL-SOA. Although SRL-SOA achieves higher MSD values, its classification accuracy is lower than our algorithm. This is because the band subset selected by SRL-SOA contains several low entropy bands, such as [104, 144], which may cause noise interference and make the MSD value abnormally higher than other algorithms.

4) Comparison Results on SA Dataset: As seen from the classification result curves in Fig. 13, our results closely match the performance of some of the latest algorithms, such as S4P, RDGSR, and DARecNet, but higher than other algorithms, especially when only a few bands are used, such as three and five bands. In addition, we found an interesting phenomenon that an algorithm is not always applicable to all different datasets, and its performance on different datasets is related to the algorithm itself or the characteristics of the data. For instance, EFDPC does not work well on the IP dataset, but it shows comparable performance on SA dataset. Across the

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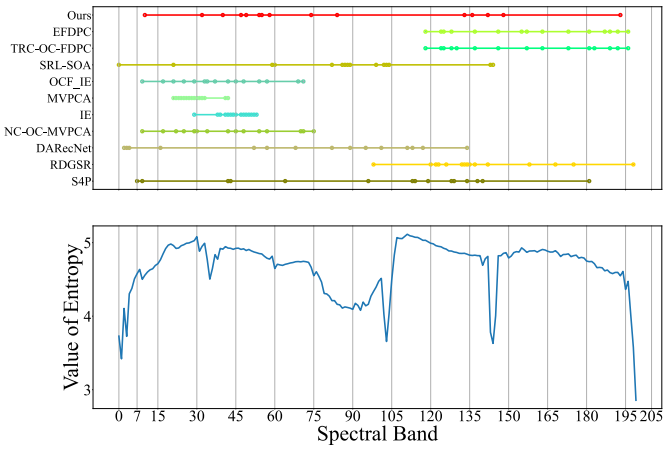


Fig. 11. Best 15 bands in top 60 salient bands of the corrected IP dataset selected by (top) different BS methods and (bottom) entropy value of the whole spectral bands.

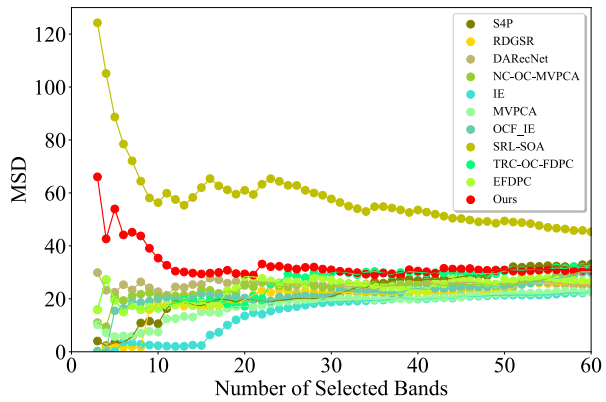


Fig. 12. Comparison of MSD under different band subsets selected by different BS methods on the corrected IP dataset.

OA, AA, or Kappa curves, the performance of different size band subsets selected by our method is comparable to that of using all bands. This obviously shows that our method can commendably obtain the saliency information of SA dataset.

The best 15 bands among the top 60 bands extensively analyzed using different methods are listed in Table II. At the same time, we mark the location of each band in the entire spectrum in Fig. 14. It can be seen from the entropy distribution map that some noise bands have very low entropy values, and we list several typical representatives from all noise bands, such as [106, 107, 146, 147, 148, 203, 204]. However, the bands selected by our method can be relatively evenly distributed in the whole spectrum and can avoid these sharply decreasing low entropy regions. In addition, from the distribution of selected bands, some BS algorithms include many continuous bands, such as OCF_IE, MVPCA, IE, and NC-OC-MVPCA. Although their selected bands exhibit high entropy values, the consecutive bands result in strong correlations, which hinder achieving significant classification performance.

Fig. 15 shows the MSD values of different BS methods. Among them, the MSD values of SRL-SOA, EFDPC, and TRC-OC-FDPC are abnormally higher than other algorithms. This is because they all contain bands with very low entropy,

TABLE I

BEST 15 BANDS IN TOP 60 SALIENT BANDS SELECTED BY DIFFERENT BS METHODS ON THE CORRECTED IP DATASET

Methods	Selected Band Subsets
Ours	84/74/10/193/133/58/148/ 40/47/136/49/54/142/55/32
EFDPC	173/128/192/157/163/125/146/ 118/181/189/187/124/155/196/137
TRC-OC-FDPC	173/128/192/157/163/125/146/ 118/181/189/124/196/137/183/130
SRL-SOA	103/0/82/59/60/86/21/ 99/88/104/89/87/102/143/144
OCF_IE	42/48/45/29/34/21/33/ 25/17/54/37/69/71/9/57
MVPCA	29/28/26/27/25/30/24/ 23/32/42/22/33/31/41/21
IE	42/41/43/44/39/48/49/ 50/51/52/45/38/53/47/29
NC-OC-MVPCA	29/25/30/42/22/34/48/ 17/45/57/54/75/70/9/71
UBS	1/14/27/41/54/67/81/ 94/107/121/134/147/161/174/187
DARecNet	57/134/16/111/95/113/89/ 117/52/101/2/3/68/82/4
RDGSR	133/132/122/120/123/134/126/ 198/175/142/98/135/168/137/158
S4P	114/128/119/134/113/43/64/ 140/129/181/7/42/138/9/96

TABLE II

BEST 15 BANDS IN TOP 60 SALIENT BANDS SELECTED BY DIFFERENT BS METHODS ON CORRECTED SA DATASET

Methods	Selected Band Subsets
Ours	103/120/6/39/157/90/67/ 185/162/104/81/139/171/77/99
EFDPC	166/68/148/88/32/131/192/ 55/177/96/185/93/199/20/82
TRC-OC-FDPC	166/68/148/88/32/131/192/ 55/177/96/93/199/20/82/121
SRL-SOA	2/1/106/0/6/202/105/ 5/203/90/107/4/164/104/25
OCF_IE	45/52/50/57/34/35/30/ 40/25/72/70/74/37/66/21
MVPCA	45/46/42/47/44/48/52/ 51/53/41/54/55/50/56/49
IE	45/46/47/42/48/52/51/ 44/55/53/54/56/50/41/49
NC-OC-MVPCA	45/52/50/57/40/32/30/ 35/72/70/74/25/66/37/21
UBS	1/14/28/41/55/69/82/ 96/109/123/137/150/164/177/191
DARecNet	68/47/17/27/51/55/76/ 8/45/77/50/93/98/71/86
RDGSR	125/89/71/86/29/72/67/ 73/87/85/129/57/102/143/116
S4P	99/35/47/48/188/164/82/ 37/2/127/200/95/149/171/21

which can be seen from the band distribution. Our method obtains MSD values similar to RDGSR, which can also be proved by their classification performance.

5) Comparison Results on PU Dataset: The performance comparison of different BS methods on the PU dataset is shown in Fig. 16, which includes the classification results of KNN and SVM. The result distribution of each algorithm in the top 30 bands is quite different, especially when fewer bands are selected, which has the same characteristics on

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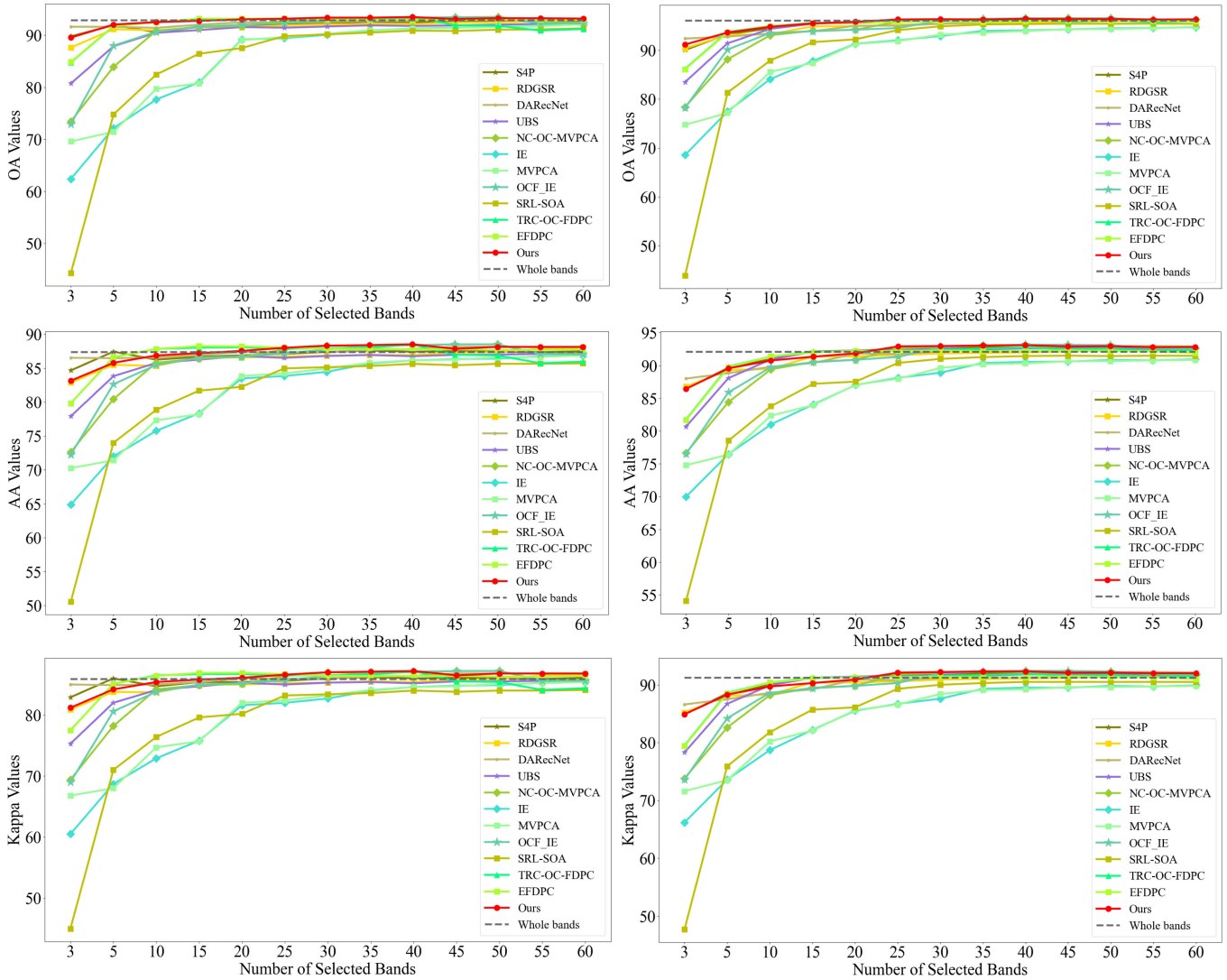


Fig. 13. Performance comparison of different BS methods on the corrected SA dataset. The figure shows the results of the top 60 bands. (Left column) Results obtained by the KNN classifier. (Right column) Results obtained by the SVM classifier. (First–third rows) OA curves, AA curves, and Kappa curves, respectively.

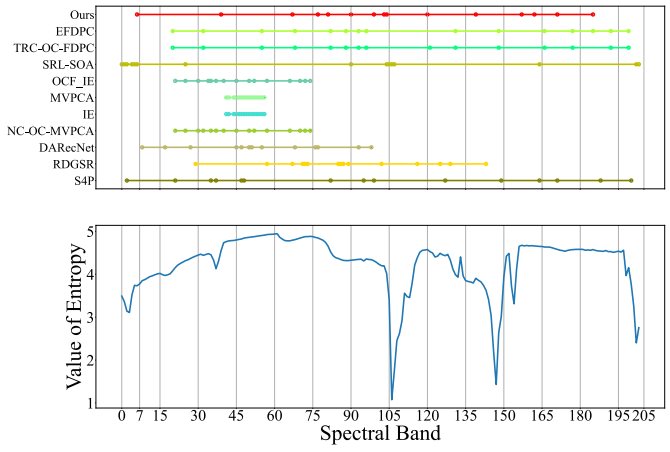


Fig. 14. Best 15 bands in top 60 salient bands of corrected SA dataset selected by (top) different BS methods and (bottom) entropy value of the whole spectral bands.

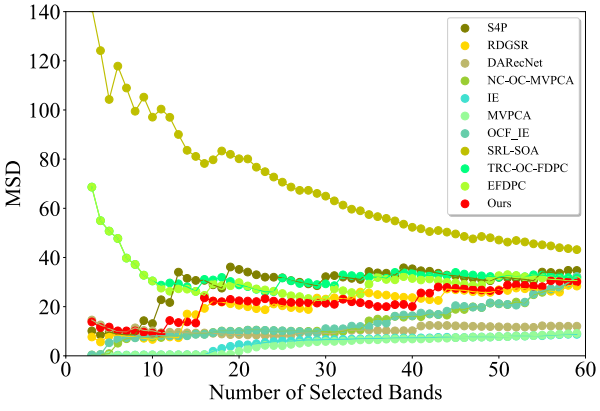


Fig. 15. Comparison of MSD under different band subsets selected by different BS methods on corrected SA dataset.

other datasets. However, our algorithm can still achieve high performance when fewer bands are selected. After exceeding the top 30 bands, the performance of all algorithms is relatively similar, which is distributed around the all bands performance curve and fluctuates slightly. The classification performance

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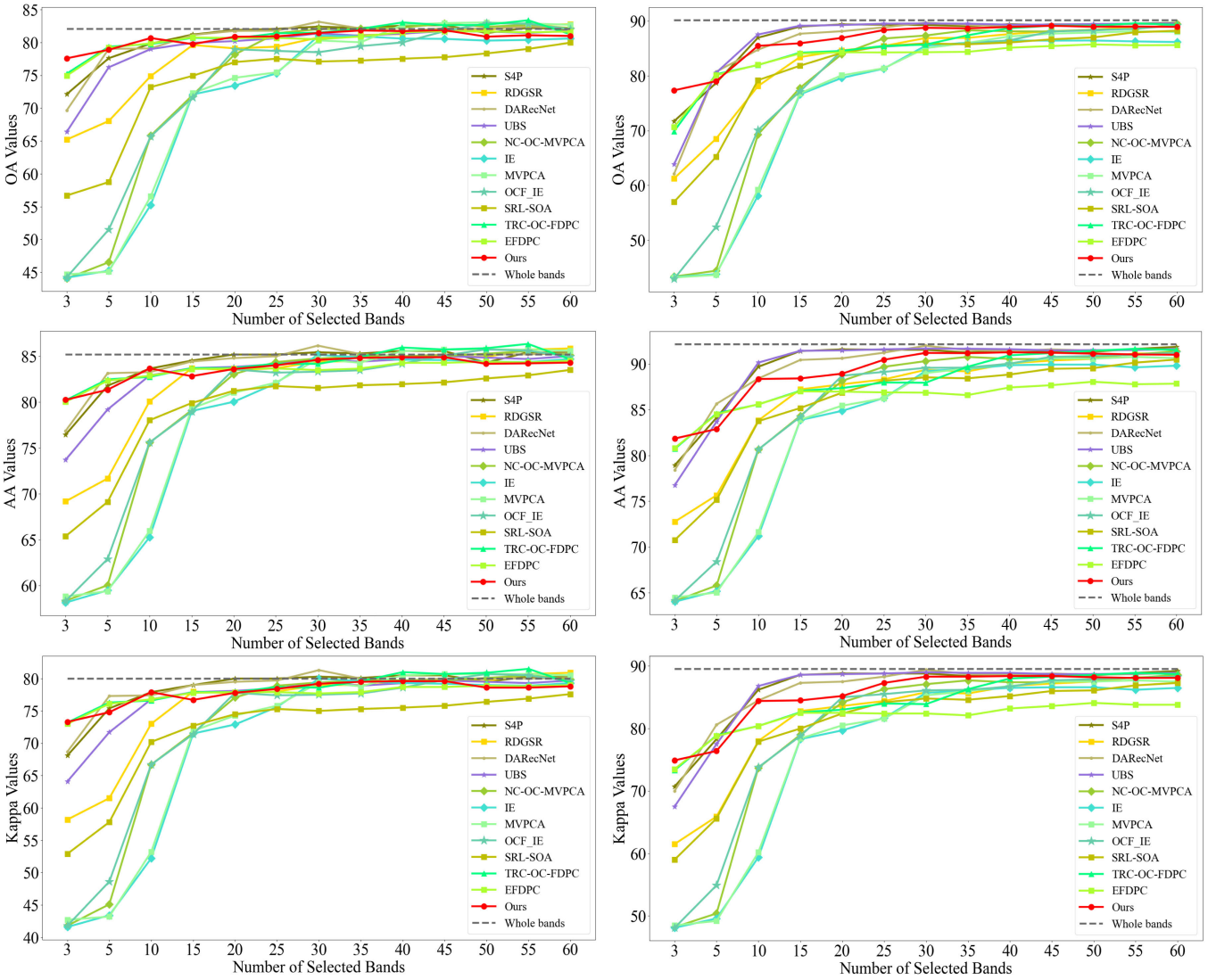


Fig. 16. Performance comparison of different BS methods on PU dataset. The figure shows the results of the top 60 bands. (Left column) Results obtained by the KNN classifier. (Right column) Results obtained by the SVM classifier. (First–third rows) OA curves, AA curves, and Kappa curves, respectively.

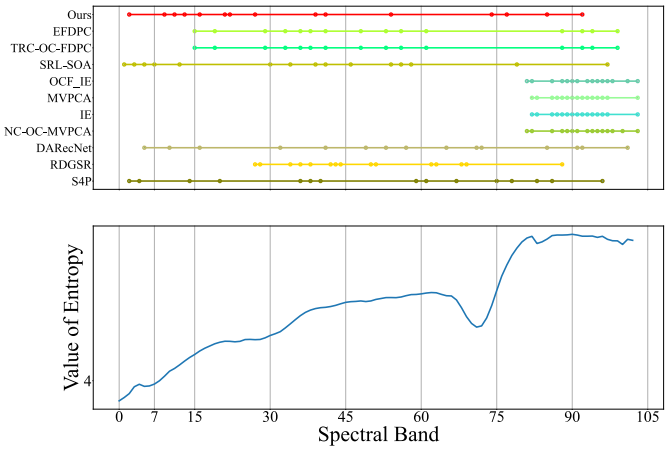


Fig. 17. Best 15 bands in top 60 salient bands of PU dataset selected by (top) different BS methods and (bottom) entropy value of the whole spectral bands.

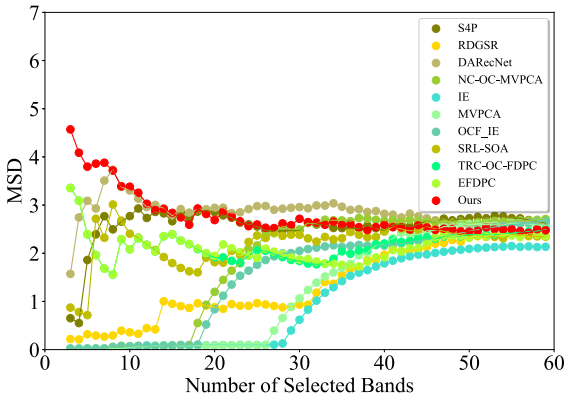


Fig. 18. Comparison of MSD under different band subsets selected by different BS methods on PU dataset.

curves sufficiently verify that our algorithm can effectively address the challenges of the PU dataset and also prove that the saliency analysis of HSI can promote the classification performance of different band subset sizes as close to or even more than the whole bands as possible.

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TABLE III
BEST 15 BANDS IN TOP 60 SALIENT BANDS SELECTED BY DIFFERENT BS METHODS ON PU DATASET

Methods	Selected Band Subsets
Ours	92/16/39/77/22/85/9/ 74/54/11/21/41/27/2/13
EFDPC	61/88/33/19/53/48/41/ 29/92/56/99/36/38/94/15
TRC-OC-FDPC	61/88/33/19/53/48/41/ 29/92/56/99/36/38/94/15
SRL-SOA	3/5/58/1/12/79/46/ 97/56/54/7/34/30/41/39
OCF_IE	91/90/88/89/95/93/94/ 96/82/86/97/103/81/98/101
MVPCA	91/88/90/89/87/92/93/ 95/94/96/82/86/83/97/103
IE	91/90/88/92/89/87/95/ 93/94/96/82/83/86/97/103
NC-OC-MVPCA	91/88/90/89/93/95/94/ 96/82/86/97/103/81/98/100
UBS	1/7/14/21/28/35/42/ 49/55/62/69/76/83/90/97
DARecNet	72/92/85/49/32/91/10/ 5/65/101/41/57/16/71/53
RDGSR	27/28/44/43/62/51/34/ 63/68/50/42/69/38/88/36
S4P	14/40/61/36/75/83/86/ 59/4/2/96/38/78/20/67

The best 15 bands among the selected 60 bands extensively analyzed by different methods are shown in Table III. In addition, Fig. 17 shows the corresponding band indices and their information entropy value of each spectral band. As we can see from the distribution, the entropy curve of PU dataset is relatively smooth, and the entropy value increases steadily with the whole spectral band, without regions falling sharply. The band subset selected by our method can be more evenly distributed on the whole spectrum, and can contain fewer continuous bands, which avoid the information redundancy brought by the relevant bands. In addition, the band subsets selected by EFDPC, TRC-OC-FDPC, SRL-SOA, DARecNet, and S4P can also be well-distributed. However, the band subsets selected by OCF_IE, MVPCA, IE, and NC-OC-MVPCA are completely concentrated and distributed between 80th and 105th bands. Although these bands have high entropy values, they are highly correlated and have less complementary information. Therefore, we can find that the distribution of band subsets is related to the classification performance. When a band subset can have the characteristics of high entropy and low correlation, its classification performance must be outstanding.

To further verify the redundancy properties of the band subsets selected by different methods, the MSD values of different sizes band subsets obtained by the above BS methods are given in Fig. 18. It can be seen that the MSD values of all methods are distributed within a small interval of $(0, 5)$, and it can be found from the entropy value distribution curve that there are rarely noise bands on PU dataset. In this case, larger MSD values indicate higher differences between bands, signifying richer complementary information and thus wider coverage of spectral information, resulting in greater diversity in the selected bands. This is consistent with the phenomenon

Algorithm 2 Automatic BS via HSA

Input: HSI $\mathbf{X} \in \mathbb{R}^{W\times H\times B}$, saliency cube $\mathbf{S} \in \mathbb{R}^{W\times H\times B}$, number of spectral bands B.

Ensure: Selected band number k_B, selected band indices $\mathcal{B}^*$.

Step 1: Compute per-band saliency and rank bands in descending order.

1: Compute the average saliency score S_b for each spectral band by averaging its saliency values over all spatial pixels (m, n): $S_b \leftarrow S(m, n, b), b \in [1, B]$;

2: Rank spectral bands in descending order based on their saliency scores to obtain the sorted band indices: NewBandIndex $\leftarrow S_b$.

Step 2: Reorder the saliency cube and HSI data.

3: Reorder the saliency cube $\mathbf{S}$ to obtain the sorted saliency cube: $\mathbf{S}' \leftarrow (\mathbf{S}, \text{NewBandIndex})$;

4: Reorder the HSI data $\mathbf{X}$ in the same way to obtain the sorted HSI cube: $\mathbf{X}' \leftarrow (\mathbf{X}, \text{NewBandIndex})$.

Step 3: Compute the sorted enhanced hyperspectral data: $\mathbf{X}'_e \leftarrow (\mathbf{S}', \mathbf{X}')$.

Step 4: Compute the average spectral response for each band: $S_{A_b} \leftarrow \mathbf{X}'_e(m, n, b)$.

Step 5: Automatic band selection.

5: Compute the gradient ΔS_{A_b} between adjacent spectral bands: $\Delta S_{A_b} \leftarrow |S_{A_{b+1}} - S_{A_b}|$;

6: Identify the spectral band index corresponding to the maximum gradient change as the optimal number of selected bands: $k_B \leftarrow argmax(\Delta S_{A_b})$.

Output: The number of selected bands k_B and the selected band indices: $\mathcal{B}^* \leftarrow \text{NewBandIndex}[1 : k_B]$.

that the MSD value of our method is higher than other methods when the number of bands is small. However, as the band subset size increases, the number of selected bands gradually increases, causing the MSD value of the band subset to stabilize over time and tend to the full-band MSD. It should be noted that selecting many bands may increase the probability of containing low-entropy noise bands, leading to introducing noise or redundant information.

6) Automatic Rule to Determine BS Number: Adaptively determining the number of selected bands is one crucial aspect of BS task. To alleviate this issue, we introduce an automatic rule based on the HSI saliency knowledge to adaptively determine the number of selected bands. For HSI, the saliency cube reflects the significance or importance of each point within the pixel-spectral cube, serving as an expressive representation of the discriminative characteristic. On the other hand, the original HSI data encapsulate the spatial–spectral information, offering a rich repository of material characteristics.

Consequently, the saliency value can be viewed as an important factor. If the saliency values are leveraged to weight the original hyperspectral data, an enhanced HSI can be obtained with a more comprehensive and profound understanding. Therefore, we can utilize the combined representation of saliency and original HSI values as the enhanced HSI to seek a criterion for selecting critical spectral bands. To this end, we compute the dot product between the saliency cube

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and the HSI cube to obtain the saliency–HSI combination as the enhanced HSI, and then subsequently sort the band index of this enhanced HSI in descending order according to the saliency value of each band. The location of the first rapid decline point can be used to automatically determine the number of selected bands, i.e., the number of selected bands can adaptively set as the sort index corresponding to the mutation jumps.

Specifically, assume that the HSA cube and HSI arranged in descending order by band average spectral saliency are S' and X', respectively. Then, we directly dot multiply them to obtain the sorted enhanced HSI, denoted as $X'_e \in \mathbb{R}^{H\times W\times B}$. And the average band value SA_b of bth spectral band in X'_e can be described as

$$\text{New Band Index} = \text{sort}_{\downarrow}(\{S_1, \ldots, S_b, \ldots, S_B\}) \quad (19)$$

$$S' = S(\text{New Band Index}) \quad (20)$$

$$X' = X(\text{New Band Index}) \quad (21)$$

$$X'_e = S' \cdot X' \quad (22)$$

$$SA_b = \frac{1}{W \times H} \sum_{m=1}^{W} \sum_{n=1}^{H} X'_e(m, n, b) \quad (23)$$

where $X'_e(m, n, b)$ represents the pixel gray value of X'_e at the spatial position (m, n) in bth spectral band. Based on this, the number of selected bands k_B can be determined by the maximum gradient of adjacent average band values

$$k_B = \underset{b}{\text{argmax}}\{(SA_{b+1} - SA_b)\}, \quad b = [1, 2, \ldots, B]. \quad (24)$$

To intuitively demonstrate the decision-making process, we illustrate the BS process in Algorithm 2. Furthermore, we also draw the continuous change curve of the average band values of the enhanced HSI after sorting and the classification results with a corresponding number of selected bands in Fig. 19. The curve has several mutational jumps under a certain number of bands. We mark in red, green, and purple the top three points in the curve where the front-to-back slope changes are the largest. In the case of the number of selected bands and the index corresponding to the mutation jumps, their classification performance is always around the highest value tested on different band subsets. In fact, this phenomenon is inevitably related to the intrinsic properties of HSI saliency. HSI saliency can highlight the most discriminative spectral bands and disregard the noisy bands in the entire HSI, which is consistent with our proposed principles of HSI saliency. Therefore, the number and index of selected bands before the mutation jump position represent the highlighted spectral bands.

On the contrary, it indicates that the bands contain noise. In summary, self-supervised HSA can provide crucial prior knowledge for HSI data processing, which can provide a simple and effective solution for determining the number of selected bands in BS task. Our S3A framework fundamentally enhances BS by dynamically evaluating the significance of spectral information. Unlike conventional methods such as UBS that require predefined subset sizes, our approach innovatively integrates HSI saliency quantification to automatically determine the optimal number of discriminative bands. This saliency-driven mechanism inherently captures the intrinsic information value of spectral features. Such automation proves critical in practical scenarios where the required band quantity remains unknown.

TABLE IV
OA RESULTS OF FOUR CLASSIFICATION METHODS BEFORE AND AFTER HSI SALIENCY ENHANCEMENT ON THREE DATASETS

	Stage	SSRN	FDSSC	DBDA	AMGCFN
IP_v1	Init.	89.13±0.81	94.75±0.64	94.74±1.79	94.16±1.10
	Enh.	**94.32±0.92**	**96.62±0.09**	**95.84±0.63**	**94.66±0.53**
	Δ	5.19	1.87	1.1	0.5
IP_v2	Init.	89.81±0.98	94.87±0.89	95.38±0.54	95.12±0.95
	Enh.	**94.11±0.15**	**97.24±0.38**	**96.43±0.46**	**95.45±0.55**
	Δ	4.3	2.37	1.05	0.33
SA	Init.	94.72±0.61	94.99±0.45	97.51±0.90	98.32±0.42
	Enh.	**95.46±0.35**	**97.61±0.31**	**97.85±0.46**	**98.60±0.41**
	Δ	0.74	2.62	0.34	0.28
PU	Init.	95.59±0.11	94.43±0.52	96.00±0.53	98.26±0.28
	Enh.	**96.62±0.44**	**97.17±0.19**	**96.84±1.26**	**98.32±0.47**
	Δ	1.03	2.74	0.84	0.06

D. Application of HSI Saliency in Classification Task

1) Experimental Setup: In order to further verify the effectiveness of HSA, we also conduct a large number of experiments in HSI downstream classification task to better illustrate the advantages and generalization performance of the proposed S3A. In the experimental implementation, the 3-D saliency cube S is used as an additional input to enhance the original HSI by directly performing elementwise multiplication. It is worth noting that we adopt exactly the same datasets and performance evaluation indicators as the BS task to ensure uniformity between different experiments.

To maintain fairness, we adopt the same training hyperparameters for the models before and after hyperspectral saliency enhancements. Specifically, during the training stage, we randomly select 3%, 0.5%, and 0.5% samples from the IP, SA, and PU datasets as training sets, respectively. Each experiment is repeated ten times, and hyperparameters such as learning rate, epoch, and batch size are consistent with the compared methods. In addition, the robustness of the algorithm is also an important indicator to consider. Hence, we verify the classification performance of the original IP dataset containing 20 noise bands to prove the ability of hyperspectral saliency to deal with redundancy and noise.

2) Classification Methods in Comparison: Specifically, in the HSI classification experiment, we compare with the following state-of-the-art methods: the spectral–spatial residual network (SSRN) (see [74]) based on 3-D CNN, the fast dense spectral–spatial convolutional network (FDSSC) [75], the DBDA [76] mechanism network, and attention multihop graph and multiscale convolutional fusion network (AMGCFN) [77]. The completely different model strategies make them suitable as representatives of the latest HSI classification methods, and the same experimental settings before and after hyperspectral saliency enhancement make the application in a fair and unbiased manner.

3) Comparison Results on Different Datasets: During the implementation of the experiment, we adopt a simple manner

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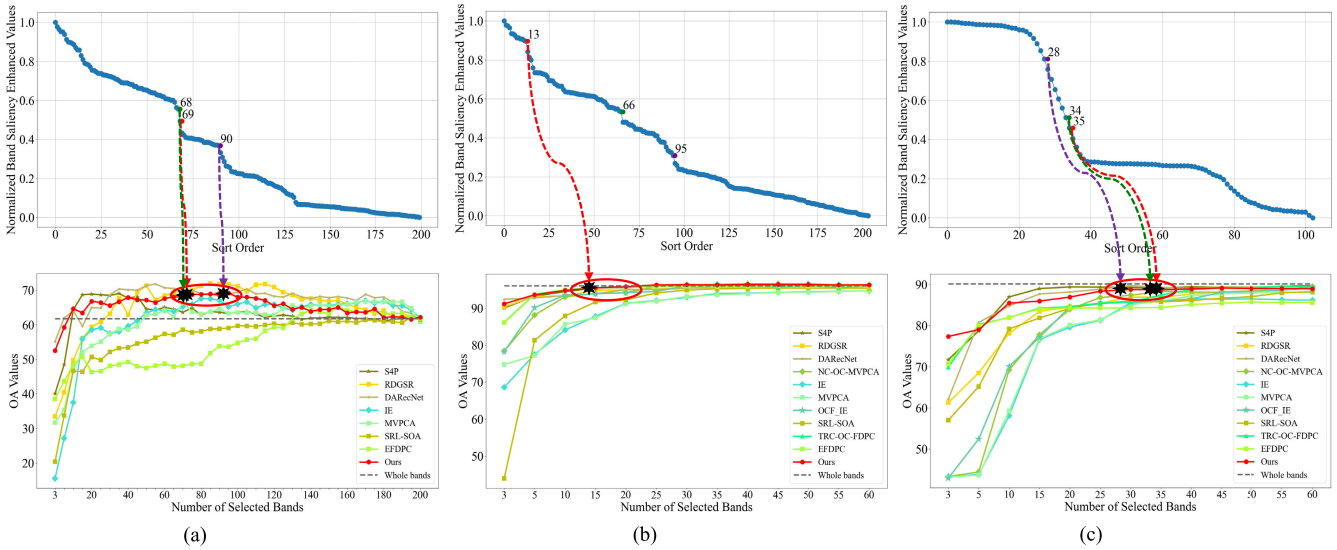


Fig. 19. Correspondence between the adaptively determined number of selected bands and the classification performance. (First row) Relationship between the sort order and band saliency-enhanced values. (Second row) OA results obtained by the SVM classifier. (First–third columns) Corrected IP, SA, and PU datasets, respectively. The top three numbers of selected bands are (a) [69, 68, 90], (b) [13, 66, 95], and (c) [35, 34, 28].

TABLE V

AA RESULTS OF FOUR CLASSIFICATION METHODS BEFORE AND AFTER HSI SALIENCY ENHANCEMENT ON THREE DATASETS

	Stage	SSRN	FDSSC	DBDA	AMGCFN
IP_v1	Init.	89.73±0.73	91.73±1.54	93.33±1.97	91.00±5.06
	Enh.	**91.17±0.77**	**94.72±0.28**	**95.24±1.69**	**93.50±1.24**
	Δ	1.44	2.99	1.91	2.5
IP_v2	Init.	79.40±2.37	94.33±1.24	96.47±2.79	93.44±2.42
	Enh.	**92.42±0.58**	**96.04±1.54**	**97.43±2.06**	**94.01±2.24**
	Δ	13.02	1.71	0.96	0.57
SA	Init.	96.66±1.08	97.56±0.94	98.00±0.79	98.77±0.41
	Enh.	**96.93±0.32**	**98.17±0.44**	**98.06±0.06**	**98.82±0.51**
	Δ	0.27	0.61	0.06	0.05
PU	Init.	94.01±0.52	94.68±0.66	96.45±0.30	96.22±0.87
	Enh.	**94.78±0.12**	**96.47±0.95**	**96.81±2.52**	**96.37±1.23**
	Δ	0.77	1.79	0.36	0.15

TABLE VI

KAPPA RESULTS OF FOUR CLASSIFICATION METHODS BEFORE AND AFTER HSI SALIENCY ENHANCEMENT ON THREE DATASETS

	Stage	SSRN	FDSSC	DBDA	AMGCFN
IP_v1	Init.	87.58±0.92	94.02±0.74	94.00±2.04	93.35±1.26
	Enh.	**93.51±1.05**	**96.15±0.09**	**95.26±0.72**	**93.92±0.60**
	Δ	5.93	2.13	1.26	0.57
IP_v2	Init.	88.39±1.13	94.14±1.01	94.74±0.61	94.44±1.09
	Enh.	**93.28±0.18**	**96.85±0.43**	**95.93±0.52**	**94.82±0.63**
	Δ	4.89	2.71	1.19	0.38
SA	Init.	94.12±0.67	94.44±0.50	97.23±0.99	98.13±0.47
	Enh.	**94.94±0.40**	**97.35±0.34**	**97.61±0.51**	**98.45±0.46**
	Δ	0.82	2.91	0.38	0.32
PU	Init.	94.15±0.14	92.57±0.67	94.67±0.71	97.69±0.37
	Enh.	**95.53±0.57**	**96.25±0.25**	**95.81±1.67**	**97.77±0.62**
	Δ	1.38	3.68	1.14	0.08

to directly multiply the HSA cube obtained by our S3A with the original hyperspectral data to enhance saliency. Then, the original HSI and the enhanced HSI are input into the model, respectively, to predict the classification performance.

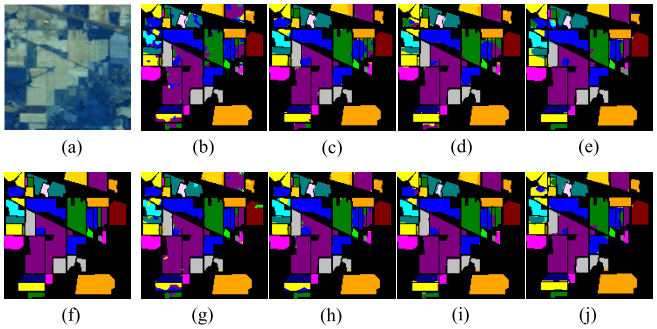


Fig. 20. Classification maps on the original IP datasets. (a) False–color image. (b)–(e) and (g)–(j) Classification results before and after HSI saliency enhancement, respectively. (f) Ground truth. Among them (b) SSRN (OA = 89.13%), (c) FDSSC (OA = 94.75%), (d) DBDA (OA = 94.74%), (e) AMGCFN (OA = 94.16%), (g) enhanced SSRN (OA = 94.32%), (h) enhanced FDSSC (OA = 96.62%), (i) enhanced DBDA (OA = 95.84%), and (j) enhanced AMGCFN (OA = 94.66%).

Tables IV–VI are the OA, AA, and Kappa performance results of four classification methods before and after HSI saliency enhancement on three datasets. Please note that "Init." and "Enh." in the tables represent the experimental status before and after HSI saliency enhancement, respectively. IP_v1 and IP_v2 represent the original and corrected IP datasets, respectively. Δ represents the increment of classification performance after saliency enhancement. It can be observed from these experimental results that the classification performance after hyperspectral saliency enhancement has been improved to varying degrees. For instance, for SSRN, which has relatively poor classification accuracy on all the original HSI datasets, the highest improvements in OA, AA, and Kappa are 5.19%, 13.02%, and 5.93%, respectively. It effectively demonstrates that the HSI data enhanced by our full-resolution saliency cube can increase the spectral discriminability of different land-cover objects and increase the relative distance between classes. Moreover, this phenomenon

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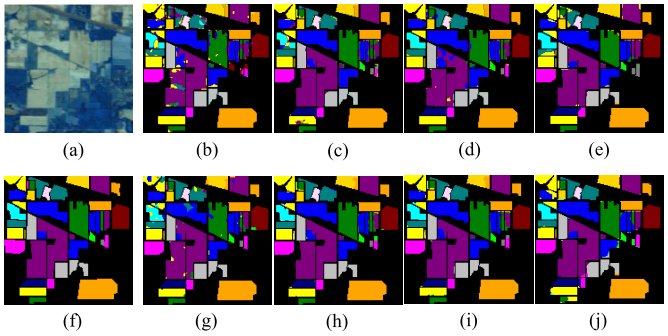


Fig. 21. Classification maps on the correct IP datasets. (a) False–color image. (b)–(e) and (g)–(j) Classification results before and after HSI saliency enhancement, respectively. (f) Ground truth. Among them (b) SSRN (OA = 89.81%), (c) FDSSC (OA = 94.87%), (d) DBDA (OA = 95.38%), (e) AMGCFN (OA = 95.12%), (g) enhanced SSRN (OA = 94.11%), (h) enhanced FDSSC (OA = 97.24%), (i) enhanced DBDA (OA = 96.43%), and (j) enhanced AMGCFN (OA = 95.45%).

is consistent with the proposed principles of HSI saliency. Specifically, as the enhancement of HSI saliency, the most discriminative spectra in the original data can be highlighted and the noise bands can be disregarded. At the same time, the saliency obtained through homogeneous region diffusion reduces the spectral differences of the same land-cover objects, which can effectively improve the overall classification performance and enable it to be effectively applied to different downstream tasks.

To visually demonstrate the improvement in the classification task after hyperspectral saliency enhancement, we visualize the classification result maps, which are illustrated in Figs. 20–23. The results show that after saliency enhancement, the homogeneous area contains fewer pixels of incorrectly predicted categories, and the overall classification result map is closer to the ground truth.

E. Ablation Experiments

1) Patch Classification Network: The patch classification network can be designed to employ different image classification models as the backbone. To verify the adaptability and robustness of our algorithm, we carry out our S3A with different image classification models, such as LeNet [78], ResNet [79], AlexNet [80], and GoogleNet [81]. Specifically, LeNet is a typical classification neural network with few layers, including three convolution layers, two pooling layers, and a fully connected layer, in which all the filters of convolution layers are 5×5. ResNet is a special deep neural network implemented by using a residual structure module, which can effectively extract the abstract semantic information of images. It can usually achieve excellent performance in the field of image classification and has become one of the mainstream models. Note that we choose ResNet with 34 layers in total. GoogleNet introduces an inception structure, which can integrate feature information of different scales, and according to 1×1 convolution kernel to realize dimension reduction mapping processing. In addition, two extra auxiliary classifiers are added to improve the classification performance.

We implement various patch classification networks to evaluate the classification performance of different bands on the

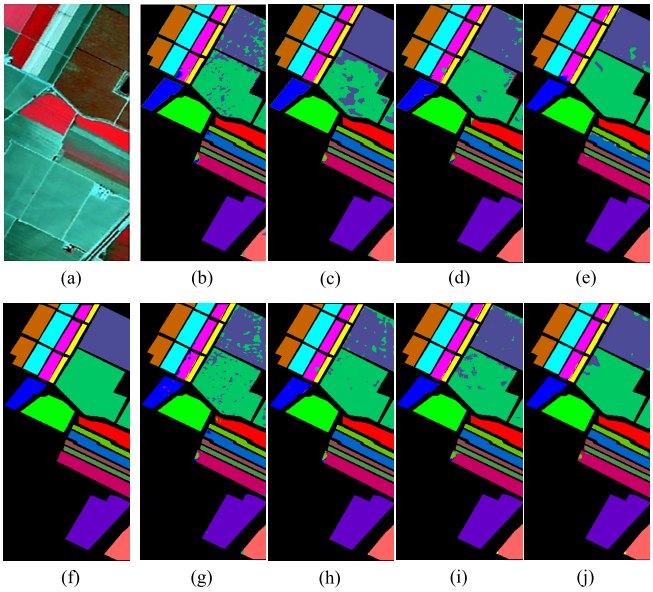


Fig. 22. Classification maps on the SA datasets. (a) False-color image. (b)–(e) and (g)–(j) Classification results before and after HSI saliency enhancement, respectively. (f) Ground truth. Among them (b) SSRN (OA = 94.72%), (c) FDSSC (OA = 94.99%), (d) DBDA (OA = 97.51%), (e) AMGCFN (OA = 98.32%), (g) enhanced SSRN (OA = 95.46%), (h) enhanced FDSSC (OA = 97.61%), (i) enhanced DBDA (OA = 97.85%), and (j) enhanced AMGCFN (OA = 98.60%).

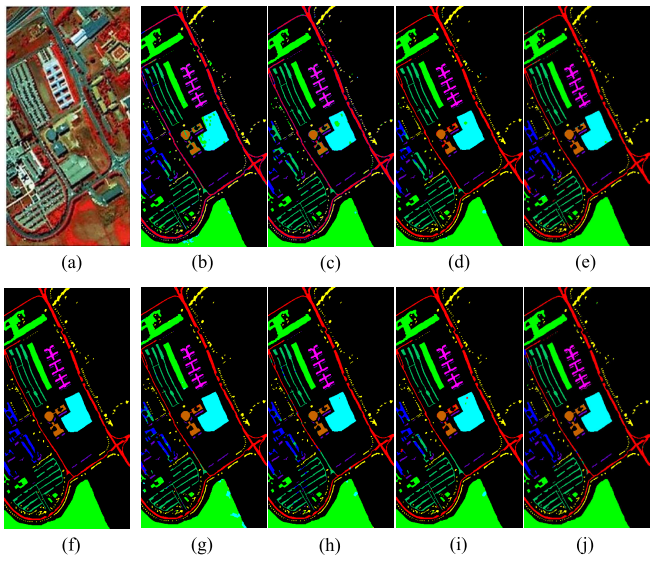


Fig. 23. Classification maps on the PU datasets. (a) False-color image. (b)–(e) and (g)–(j) Classification results before and after HSI saliency enhancement, respectively. (f) Ground truth. Among them (b) SSRN (OA = 95.59%), (c) FDSSC (OA = 94.43%), (d) DBDA (OA = 96.00%), (e) AMGCFN (OA = 98.26%), (g) enhanced SSRN (OA = 96.62%), (h) enhanced FDSSC (OA = 97.17%), (i) enhanced DBDA (OA = 96.84%), and (j) enhanced AMGCFN (OA = 98.32%).

IP dataset, using CNN-2D [82] as the classifier. As shown in Fig. 24, the curves of ResNet and Alexnet are intertwined, and both can achieve excellent performance. In addition, the OA curve of LeNet is slightly lower than that of ResNet and AlexNet, which may be due to the difficulty in extracting more reliable features from shallow networks. For GoogleNet, the large receptive field resulting from its deep layers is

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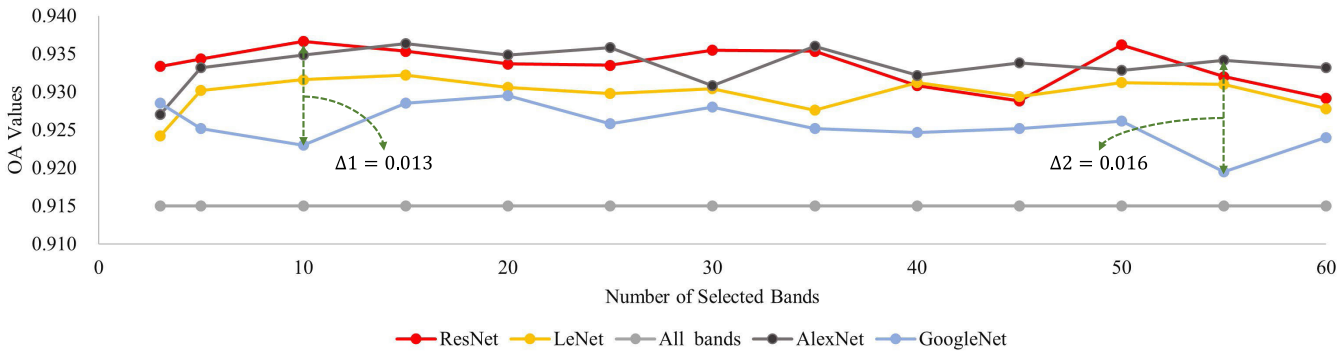


Fig. 24. Classification performance verification of our algorithm uses different image classification model as the backbone of the patch classification network on the original IP dataset.

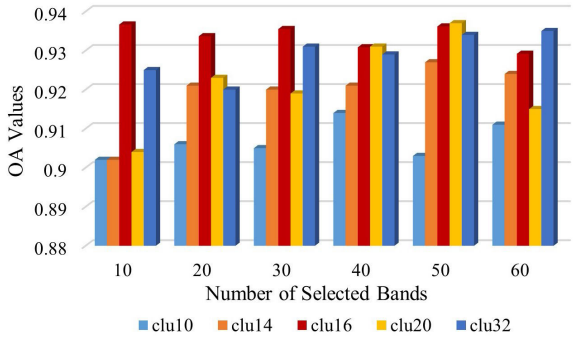


Fig. 25. Classification performance of different clusters with different band subset sizes on the original IP dataset.

disadvantageous for small input image patches. Regardless of the backbone classification network used, the OA result of our method surpasses that achieved with the full bands.

2) Cluster Numbers: Clustering land cover objects with unique spectral information can provide pseudo-label information for self-supervised HSI processing, but the number of clusters significantly affects the accuracy of downstream tasks. In the experiment, we evaluate the effect of different cluster numbers on the BS performance. Specifically, we perform a total of five different clustering experiments revolved around the real class number 16 on the original IP dataset, including [10, 14, 16, 20, 32], respectively. Meanwhile, we also verify the performance of different cluster numbers in the classification task with different band subset sizes, to conveniently reveal the essential regular patterns and fully explain the importance of cluster numbers.

As shown in Fig. 25, the classification performance is evaluated for different cluster numbers across band subset sizes ranging from 10 to 60, with ten-band intervals. It can be observed that when K is close to the actual number of pixel classes (i.e., material classes involving in the HSI data), the classification performance generally reaches the best. Therefore, K can be chosen based on the general number of categories contained in the HSI. Moreover, the overall results between different cluster numbers with the same band subset size roughly conform to the normal distribution. For instance, when the selected band subset size is 20, the results of OA under different cluster numbers are [0.906, 0.921, 0.938, 0.923, 0.92], respectively. The optimal result is obtained when the number of clusters is 16, which is exactly equal to the number of real land-cover classes. Moreover, when the cluster number is close to the real land-cover classes or integer multiples of the real classes, its performance will be improved to a greater extent. In general, the cluster numbers will affect the performance of BS, so a suitable cluster number can make our algorithm achieve better performance.

TABLE VII
AVERAGE CLASSIFICATION PERFORMANCE AND TIME CONSUMPTION UNDER DIFFERENT TRAINING DATA PROPORTIONS

IP	Elapsed time (s)	Average OA (%)	Average Kappa (%)
5%	181.04	90.3±2.6	89.0±3.0
10%	269.18	92.0±0.3	90.8±0.3
15%	354.72	92.9±0.5	91.9±0.6
20%	694.85	**93.4±0.3**	**92.4±0.3**

3) Analysis of Training Data Proportion and Time Consumption: According to the saliency of HSI to complete BS, it is necessary to calculate the importance of each pixel in the spectral dimension, which is usually used as the front-end operation of data processing in practical applications. Therefore, the key problem is how to quickly calculate the saliency per pixel and effectively reduce unnecessary time consumption while maintaining high accuracy. In this article, we design a multidimensional nonlinear filtering method by analyzing the characteristics of the data itself, which can quickly generate the significance of the global pixels when obtaining a part of the data weights. However, the accuracy of band significance is directly related to the number of samples participating in training. We carried out comparative experiments under different training and diffusion data ratios to explore the best balance between high accuracy and low time consumption.

Table VII presents the comparison results of classification accuracy and time consumption when we use different proportions of training data. Note that average OA and average Kappa represent the average classification accuracy under different band subset sizes, respectively, which helps to understand the overall classification performance of the algorithm more comprehensively. As we can see, the average OA and the average Kappa are rising with the increase of the data involved

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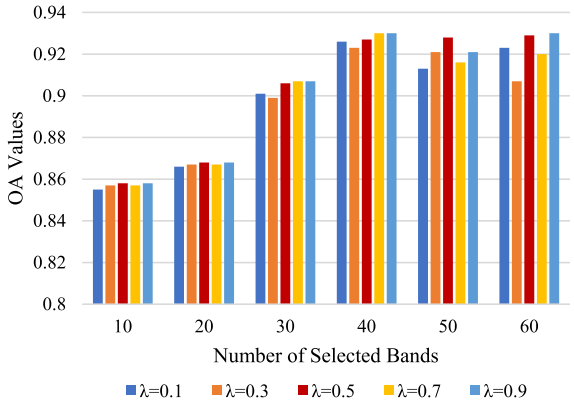


Fig. 26. Classification performance of different λ values with different band subset sizes on the original IP dataset.

in the training process. When 20% of the data participate in the training process, the average OA and the average Kappa can reach 93.4% and 92.4%, respectively, and each epoch consumes about 694.85 s. Therefore, our S3A achieves a relatively appropriate balance in terms of time cost and AA.

4) Tradeoff Parameter λ: The tradeoff parameter λ in the total dissemination function is an important factor to balance the spatial and spectral information, which are the two most critical components in hyperspectral data processing. In many hyperspectral tasks, a similar tradeoff hyperparameter λ has been mentioned [83], [84], and different λ values may cause a certain impact on the task performance. Generally, λ is set to 0.5, which is a simple and intuitive choice to reflect equal importance between spatial and spectral information. Moreover, in the process of HSI saliency diffusion, the contribution of spatial and spectral information to the objective function should have a similar degree of importance.

To further verify the impact of λ on model performance, we conducted sensitivity testing experiments for different λ values with [0.1, 0.3, 0.5, 0.7, 0.9], respectively, and performed classification experiments on the original IP dataset with band subset sizes of [10, 20, 30, 40, 50, 60]. The obtained OA results are presented in Fig. 26. Note that to reduce the time cost of model training, we randomly select 5% of the samples for S3A model learning, and the remaining samples are used for saliency diffusion. The experimental results show that different λ values have a slight impact on the performance of hyperspectral saliency, but from the overall performance, the trend of $\lambda = 0.5$ is relatively stable, which also proves to a certain extent that setting λ to 0.5 is reasonable.

V. Conclusion

In this work, we clearly defined the notion of HSI saliency through developing several principles and devised a novel S3A framework, which can mine the essential characteristics of hyperspectral data. The architecture of our S3A framework comprises three main components, including the RPS module, CPSD-Net, and P2G-SD module, which are implemented in a collaborative manner to accomplish end-to-end training. In addition, we successfully apply it to downstream tasks including BS and HSI classification on three popular HSI datasets. All extensive experiments have demonstrated the feasibility of our proposed S3A framework and its effectiveness in downstream applications, as a preliminary exploration of HSA, particularly in a condition that the concepts and principles are underdeveloped, but our S3A framework has limitations that warrant further improvement. Specifically, these include: 1) model adaptability: enhancing the model's generalization ability by incorporating an adaptive cluster number determination mechanism and leveraging parameter optimization methods to improve adaptability in scenarios with unknown class hierarchies and 2) model performance: employing more advanced model architectures and efficient collaborative strategies to accelerate the generation of more accurate 3-D saliency cubes. As a future work, we would like to investigate self-supervised HSA based on spatial–spectral union dual-stream architecture to improve the representation of HSI.

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